AUTO–ID LABS

Inside Tags

Hao Min
Director
Auto-ID Labs at Fudan

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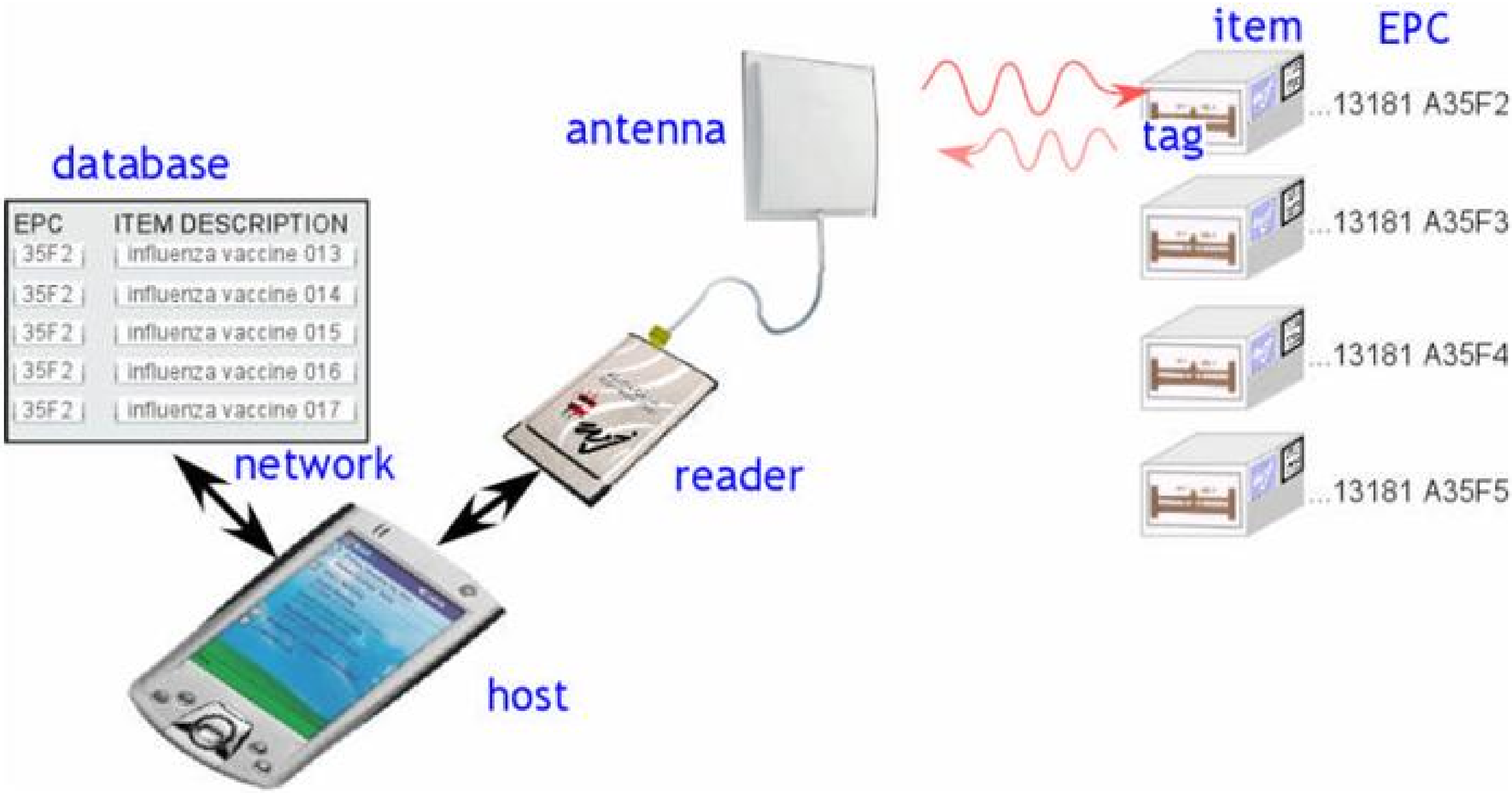
Agenda

- **Tag introduction**
- **Tag performance**
- **Tag structure**
  - Antenna
  - Chip
- **Gen 2 technology**
- **Q&A**

Introduction

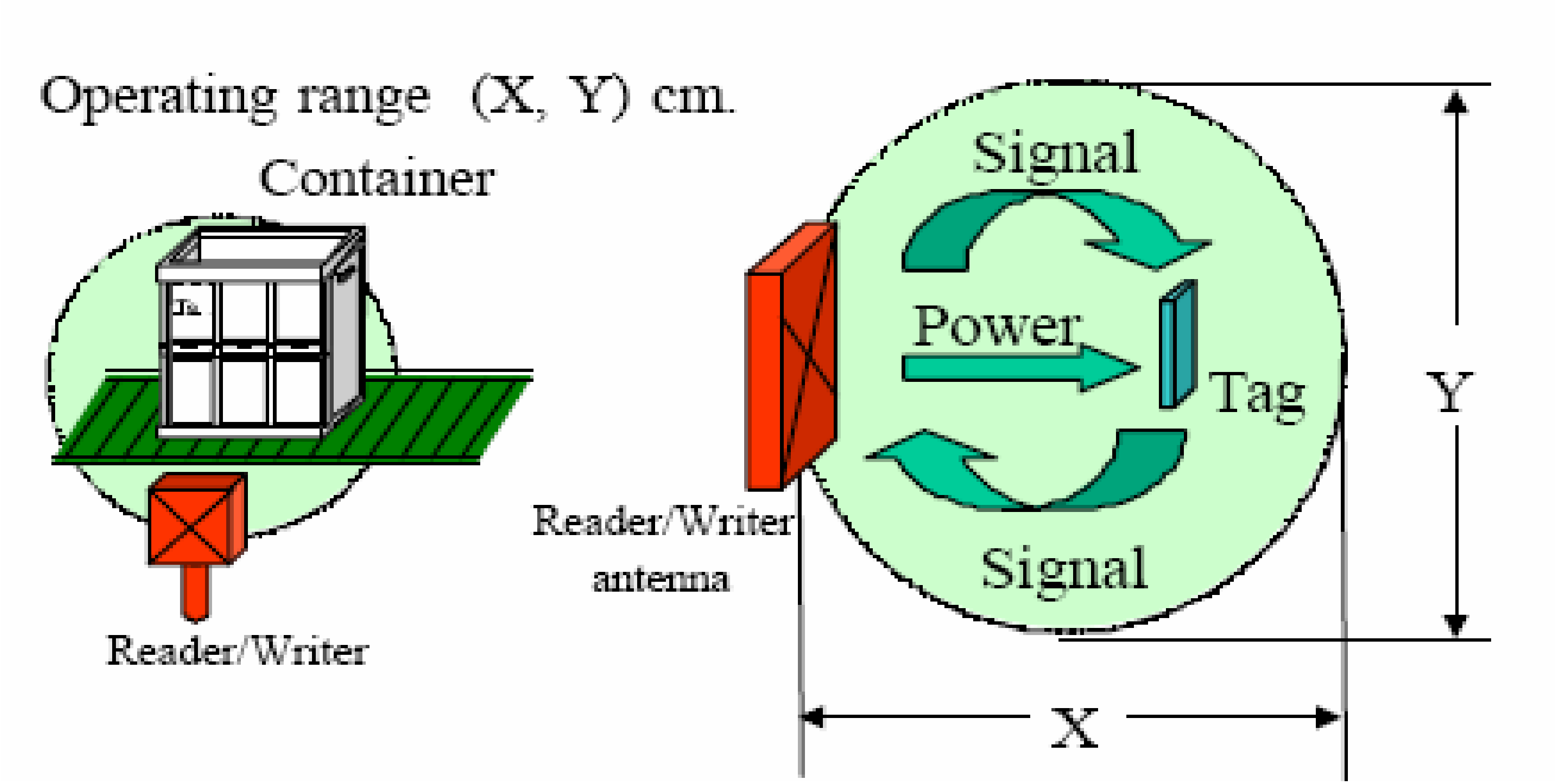
- **RFID system**
- **Frequency used in RFID**
- **Air interface**
  - Communication protocol
  - Anti-collision
  - Security
  - Down link modulation
  - Up link modulation

RFID System

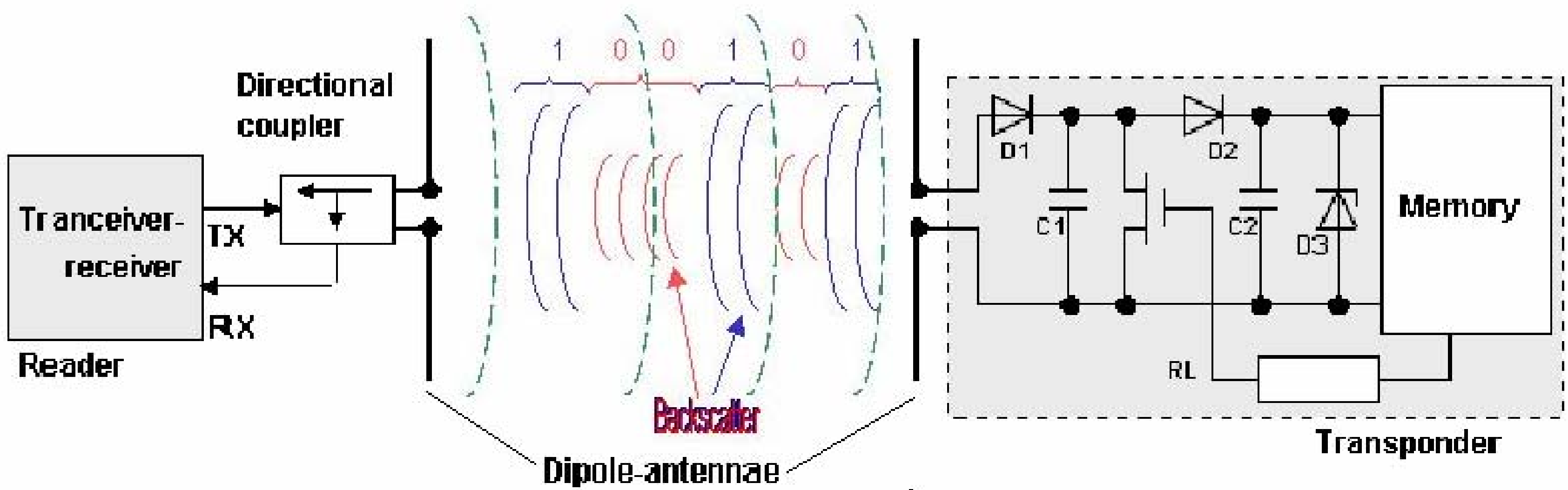


item
EPC
antenna
database
EPC ITEM DESCRIPTION
35F2 influenza vaccine 013
35F2 influenza vaccine 014
35F2 influenza vaccine 015
35F2 influenza vaccine 016
35F2 influenza vaccine 017
tag
...13181 A35F2
...13181 A35F3
...13181 A35F4
...13181 A35F5
network
reader
host

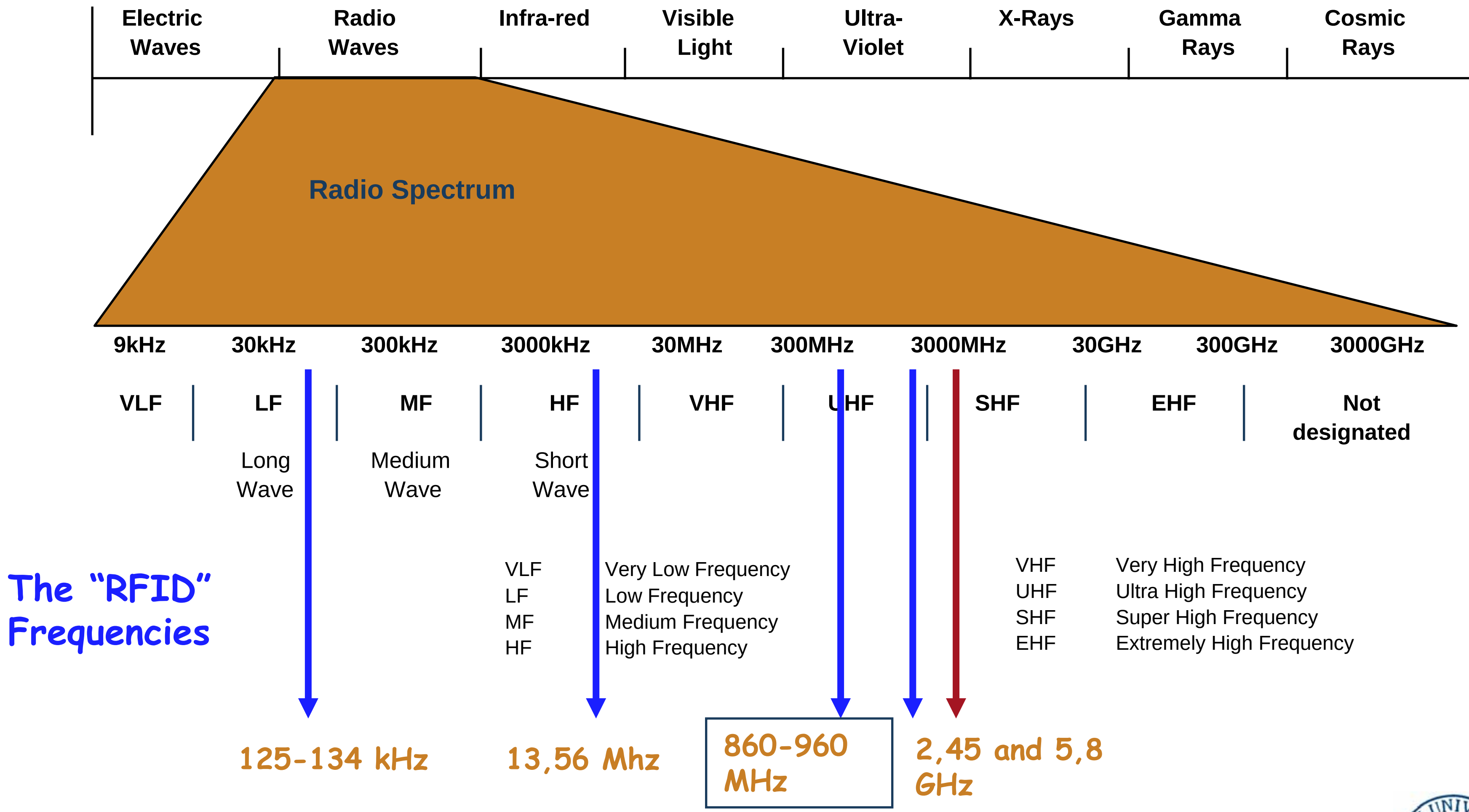
Reader and Tag Communication



Reader and Tag Communication

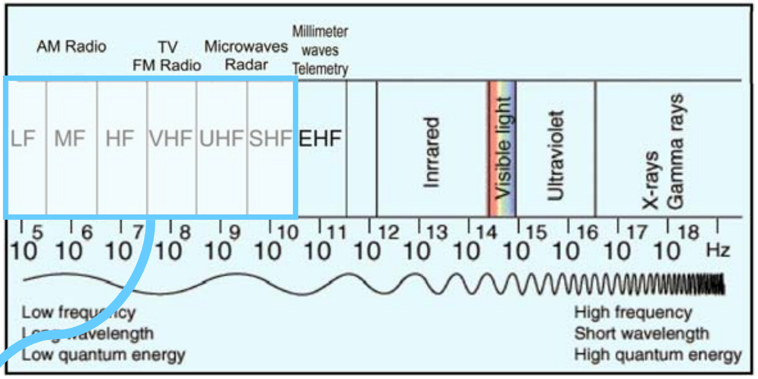


Electromagnetic Spectrum



Bands available for RFID

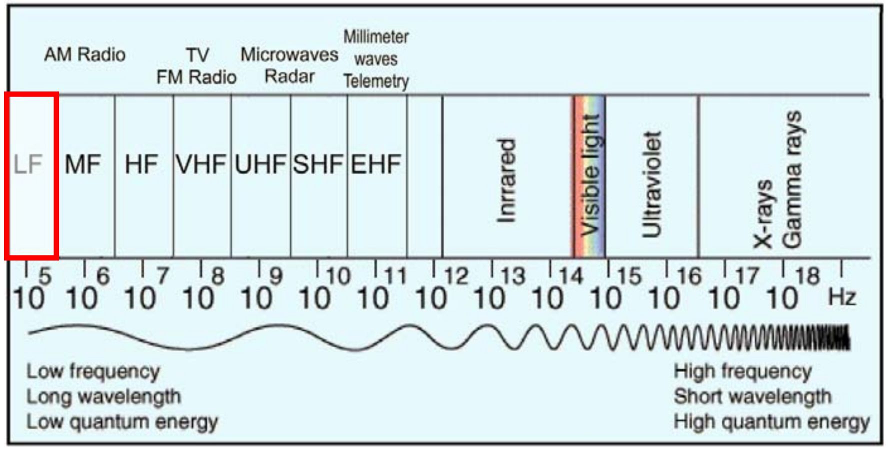
- **LF, HF and UHF allocations in most regions of the World**



Band designation	LF low frequency	MF medium frequency	HF high frequency	VHF very high frequency	UHF ultra high frequency	SHF super high frequency
Frequency	30–300kHz	300kHz–3MHz	3–30MHz	30–300MHz	300MHz–3GHz	3–30GHz
Wavelength	10–1km	1000–100m	100–10m	10m–1m	1m–0.1m	0.1–0.01m

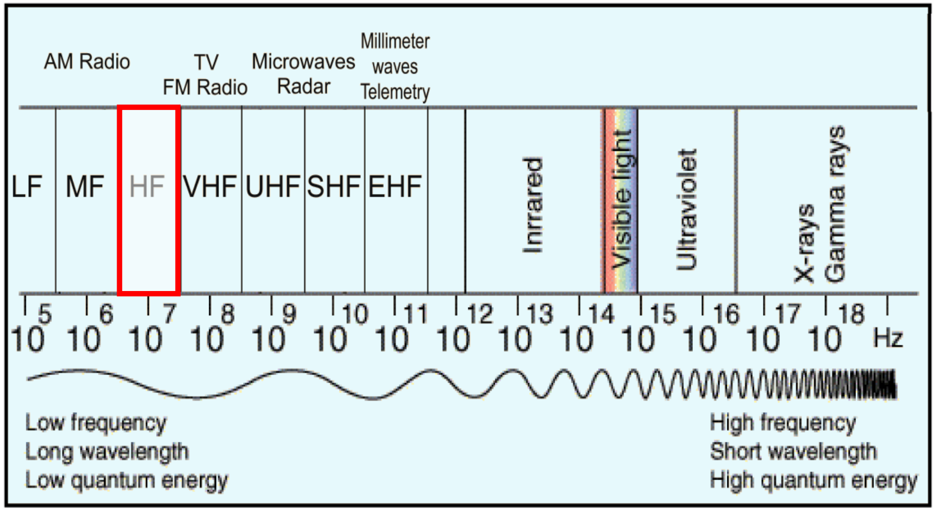
LF band

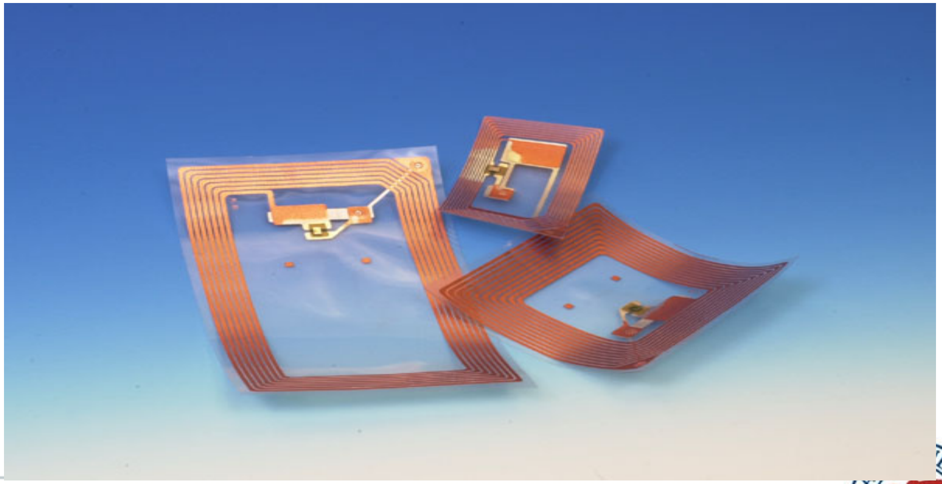
- **125kHz, 134kHz**
- **Near field coupling**
- **Relatively expensive**
- **Low range**
- **Robust, but bulky tags**
- **Widely used in**
  - Animal tracking
  - Access control



HF band

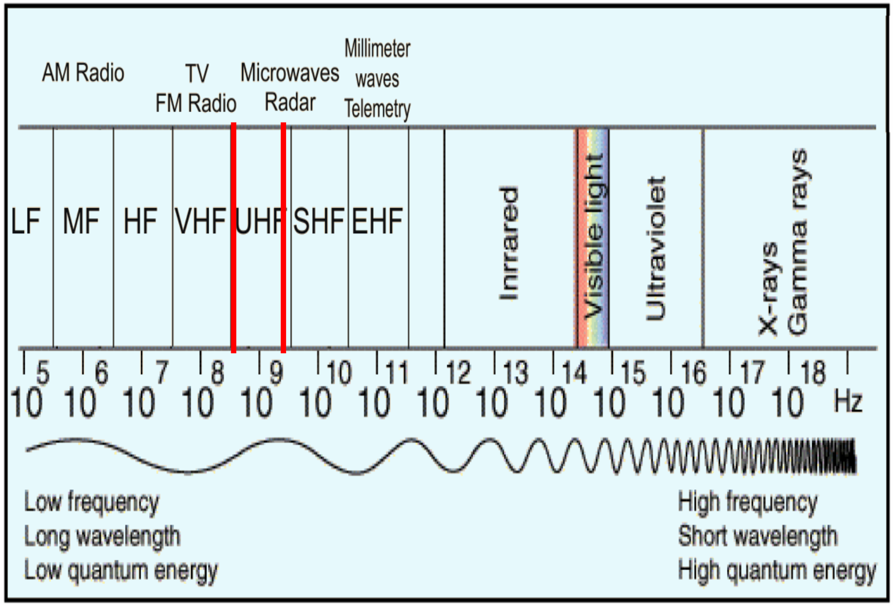
- 13.56MHz
- Near field coupling
- Read range 0.1~1m
- Reasonable cost and performance
- Discreet tags, still quite low range
- Used in access control and supply chain management





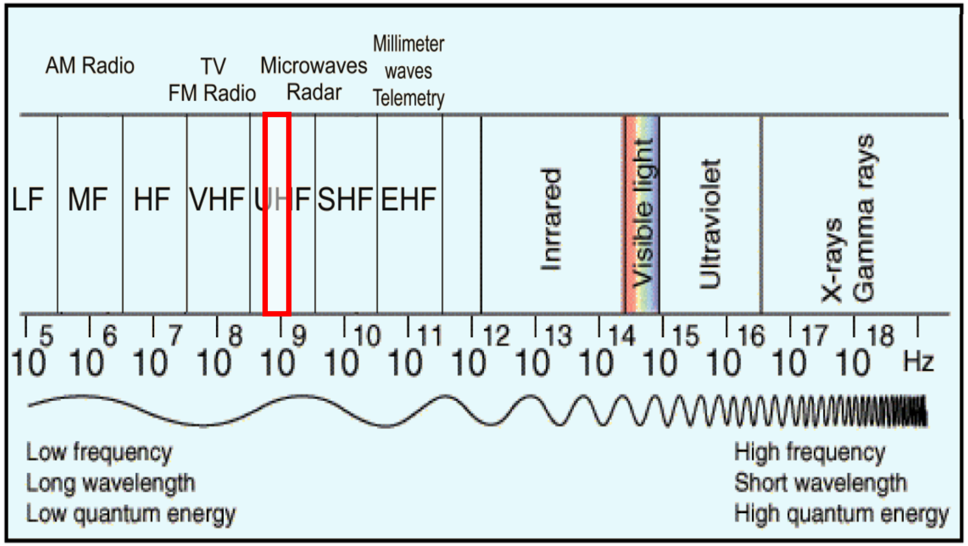
UHF bands 2.45G and 433MHz

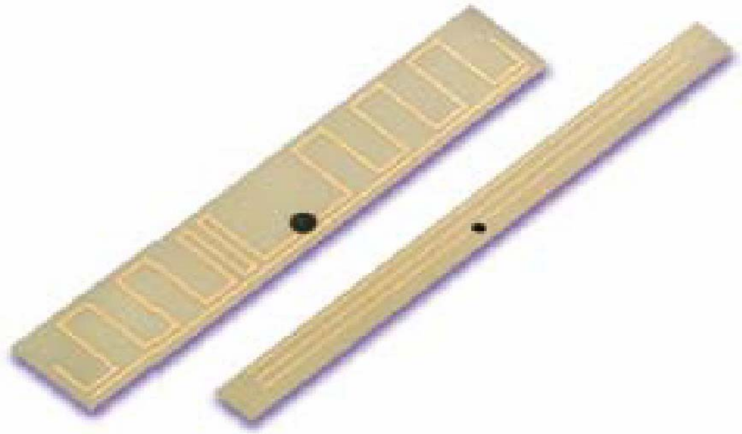
- **2.45GHz**
  - Technically also UHF, also ‘microwave’
  - Smaller antennas (and tags)
  - Typically less range ~1m
- **433MHz (and others)**
  - Specific, not really standardised
  - Used for container tagging by Savi



UHF bands 860 ~ 960 MHz

- Latest technology
- Legislation varies between regions
- Far field wave propagation and backscattering
- Read range 1 ~ 10m
- Widely used in identification and supply chain management

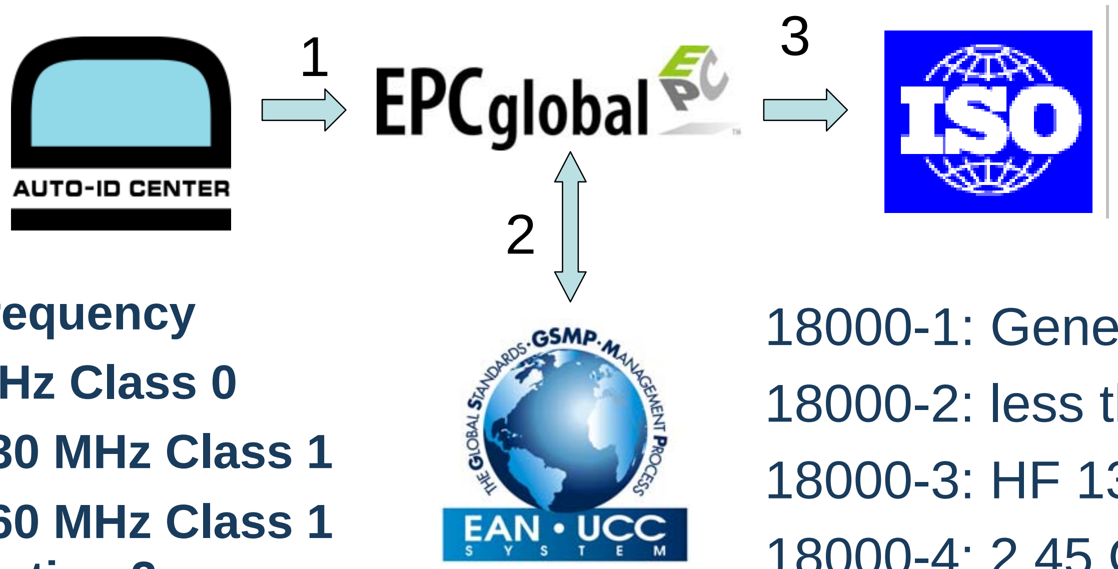




Air Interface

- Standard
- Communication protocol
- Anti-collision
- Down link modulation
- Up link modulation

Standard



Ultra-High Frequency

- **900 MHz Class 0**
- **860-930 MHz Class 1**
- **860-960 MHz Class 1 Generation 2**

High Frequency

- **13,56 MHz Class 1**

18000-1: General guidelines

18000-2: less than 135 KHz

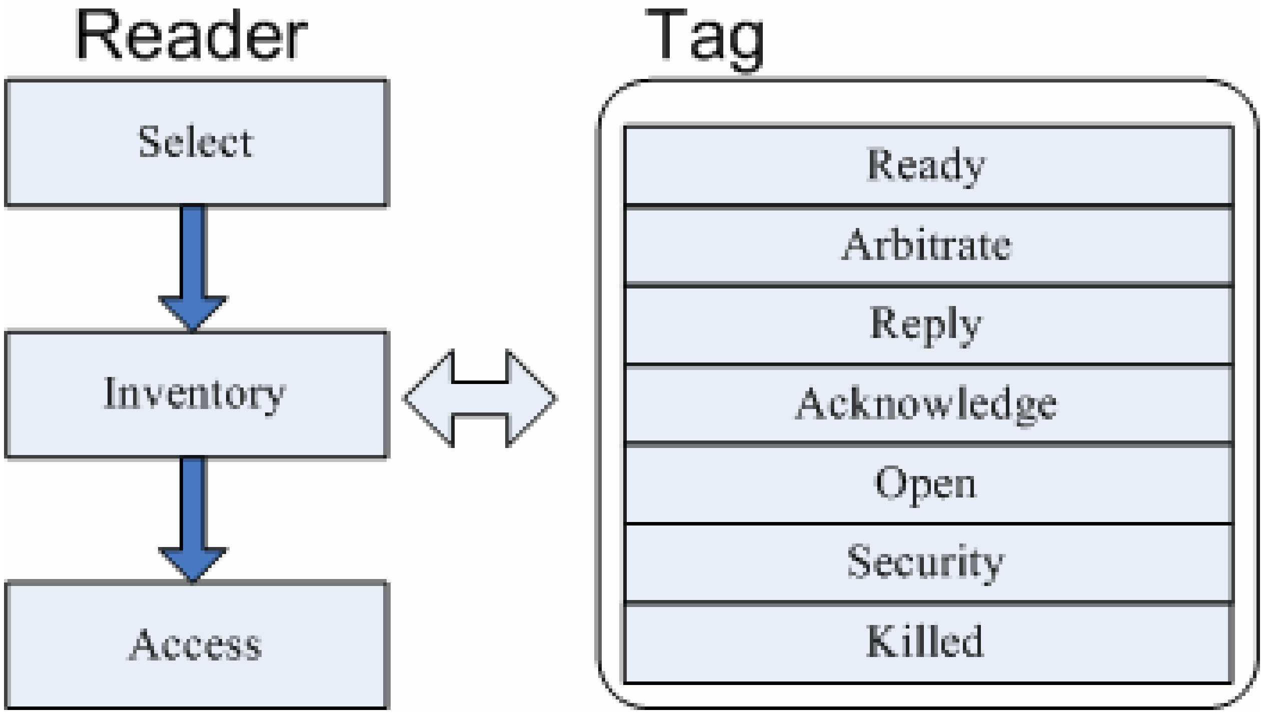
18000-3: HF 13.56 MHz

18000-4: 2.45 GHz

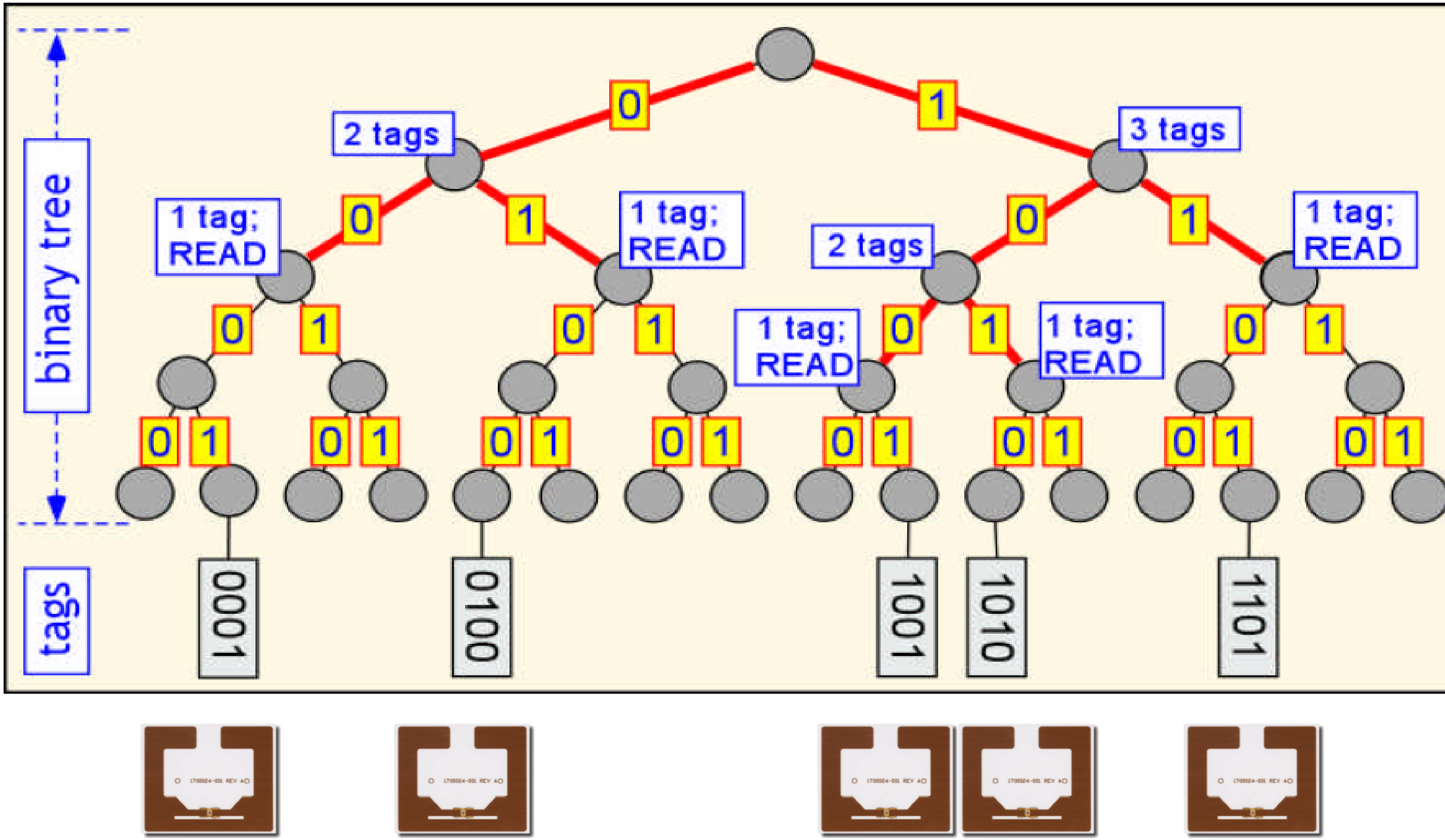
18000-6: UHF 860-960 MHz

18000-7: 433 MHz

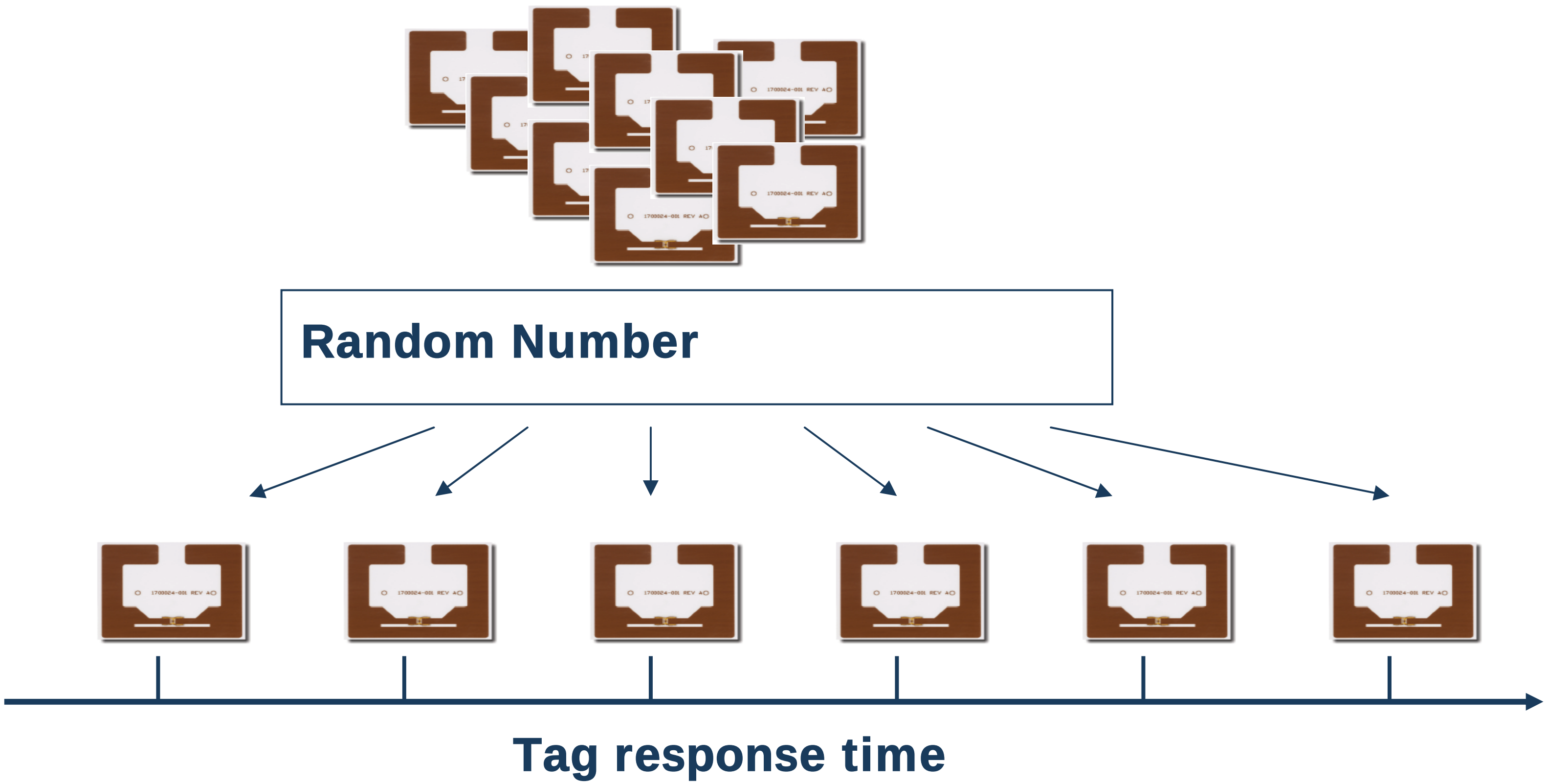
Communication Protocol



Anti-collision
Binary tree

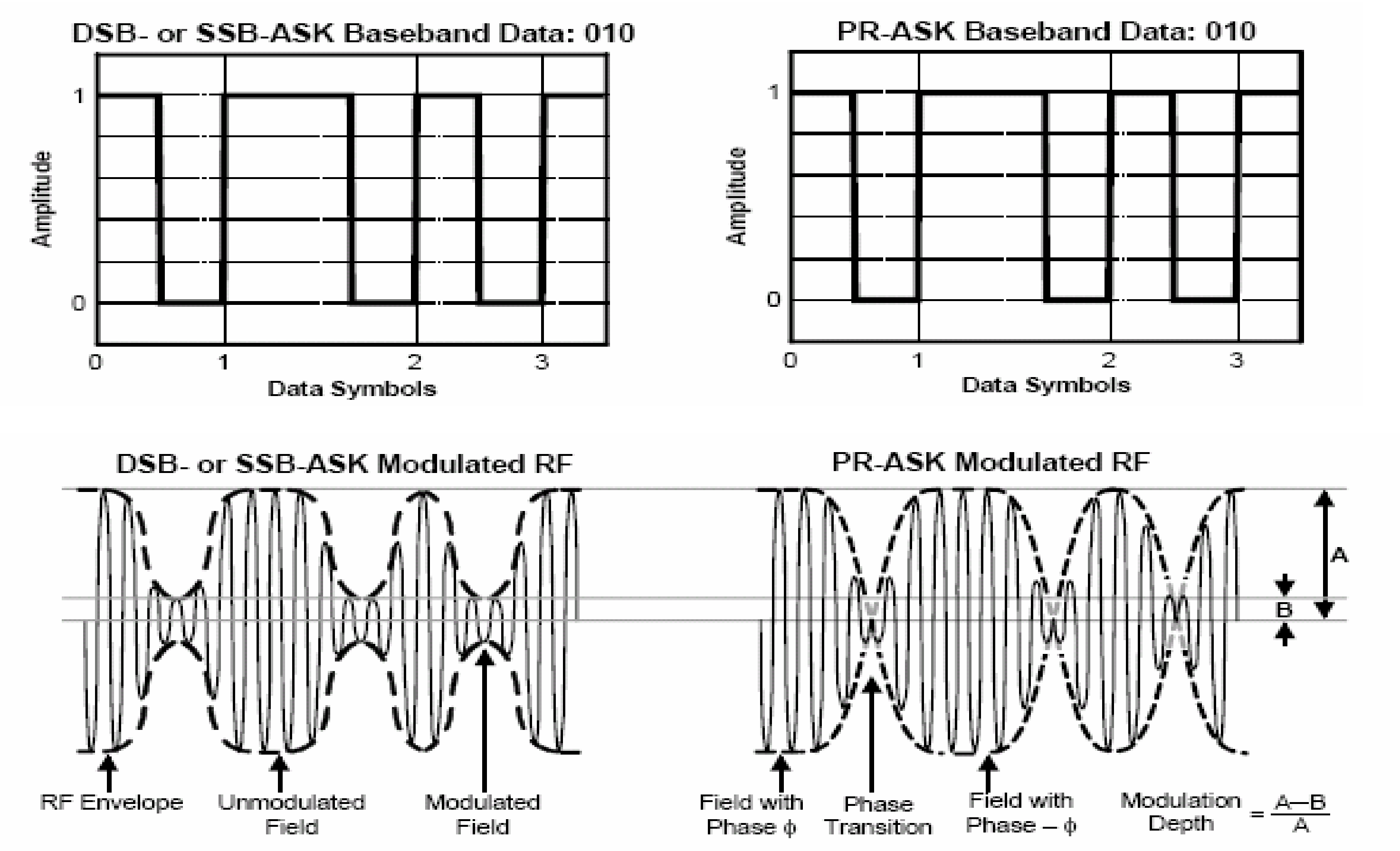


Anti-collision
Aloha



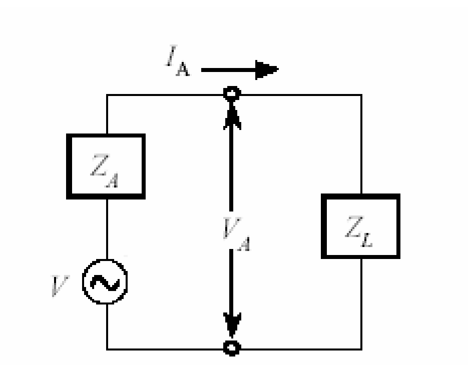
Reader to Tag Modulation

- **Amplitude modulation**



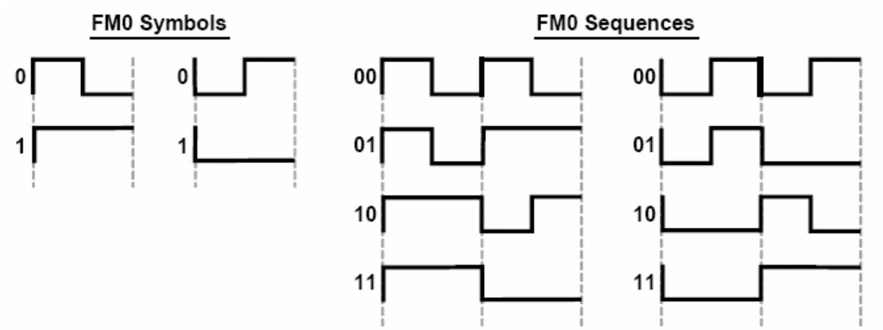
Tag to Reader Modulation

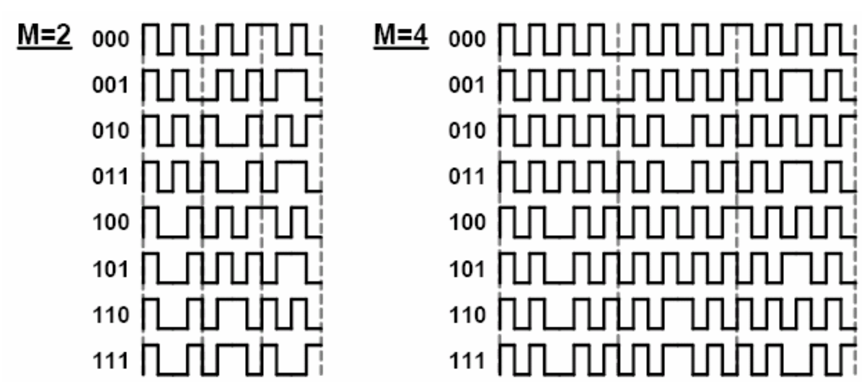
- Load modulation (HF and LF) or backscattering (UHF)
- Amplitude modulation of phase modulation
- Various coding



Equivalent circuit of a far-field tag

$$\Gamma = \frac{Z_A^* - Z_L}{Z_A + Z_L}$$





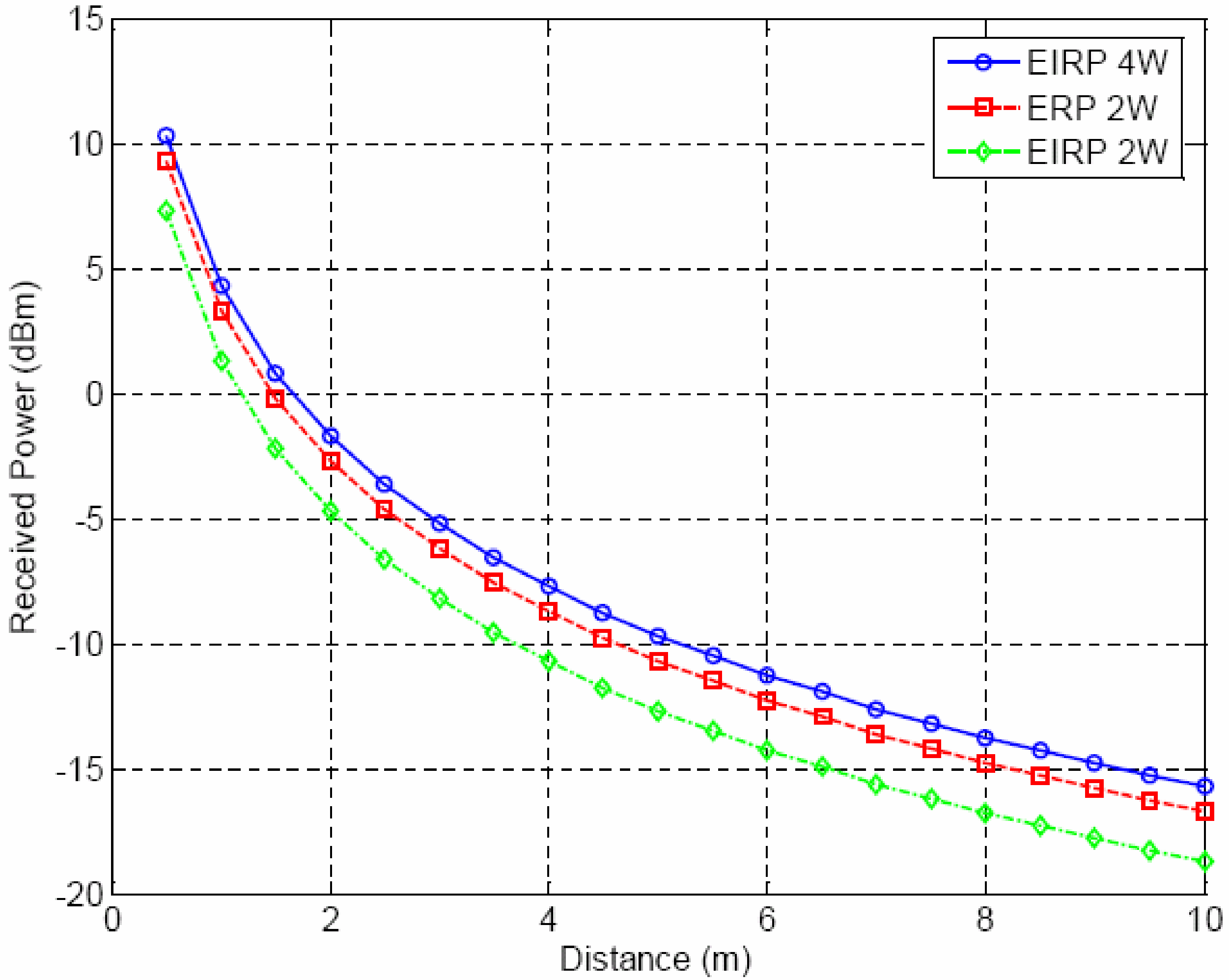
Tag performance

- Classes of tag
- Read range (minimum operating power)
- Singulate rate
- Memory
- Bandwidth

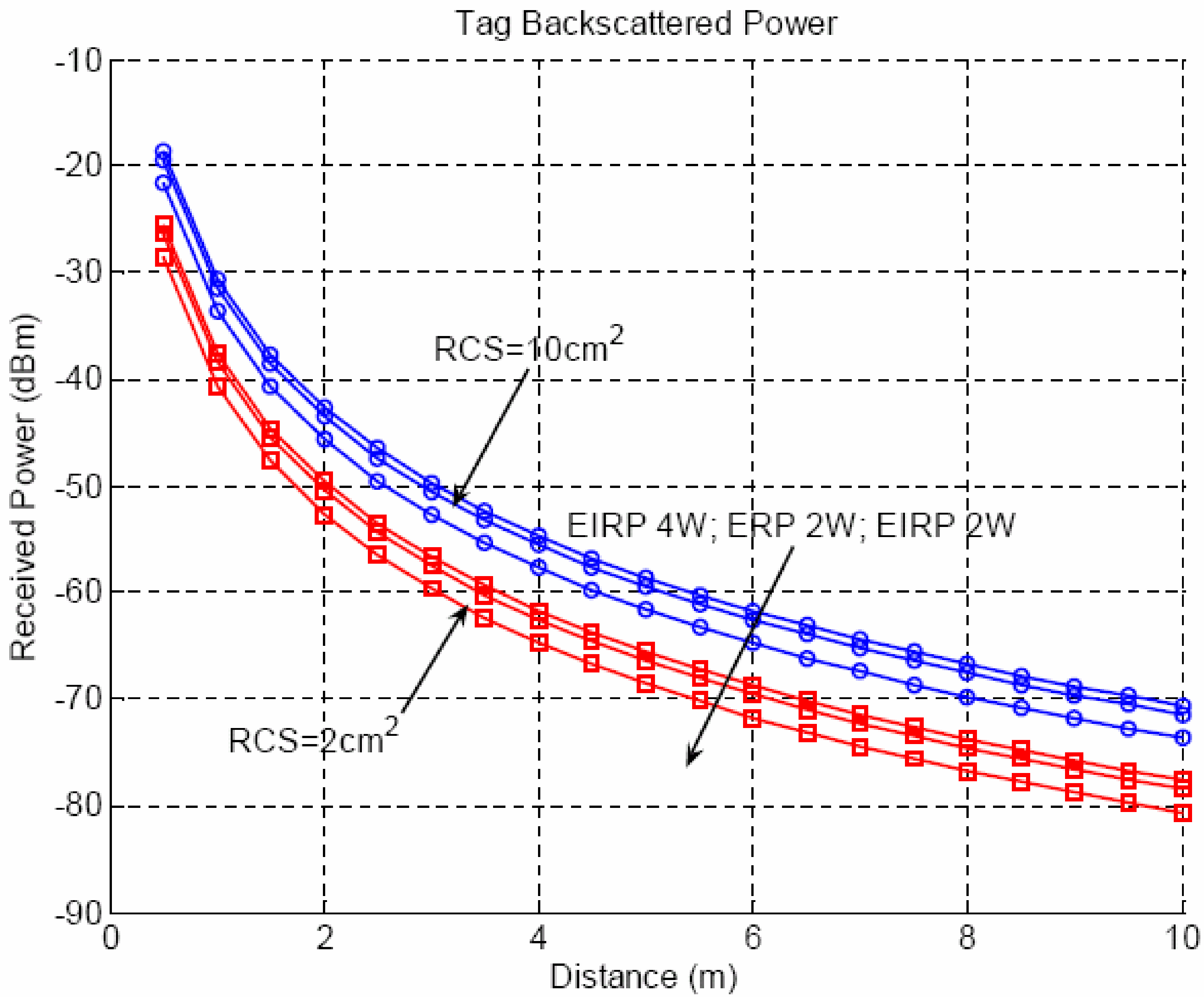
Classes of Tag

- Class 0: Read only
- Class 1: Write once
- Class 2: Read and write
- Class 3: Semi-passive
- Class 4: Active
- Class 5: Active with reader function

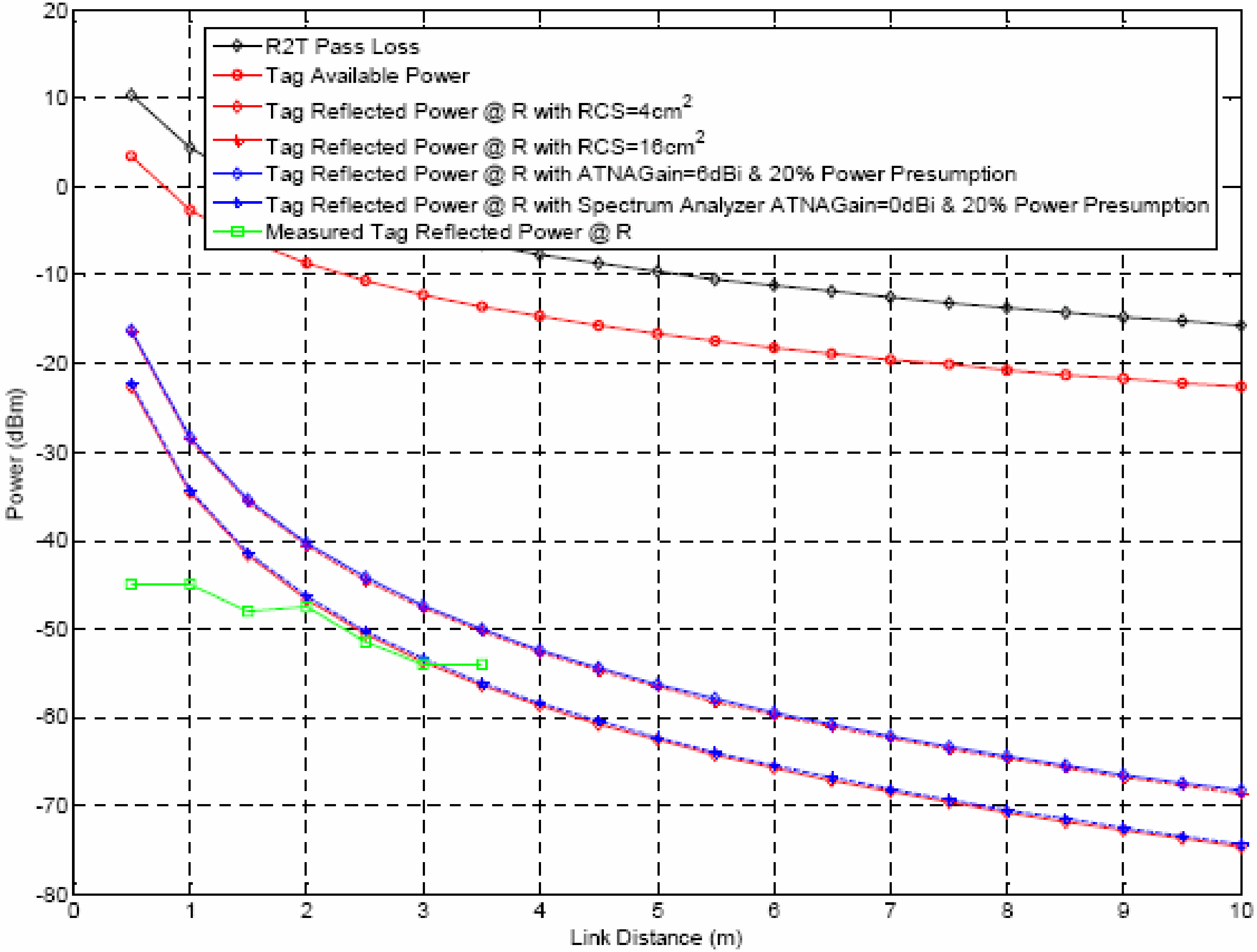
Read Range
Tag received power



Read Range
Tag Scattered Power



Read Range

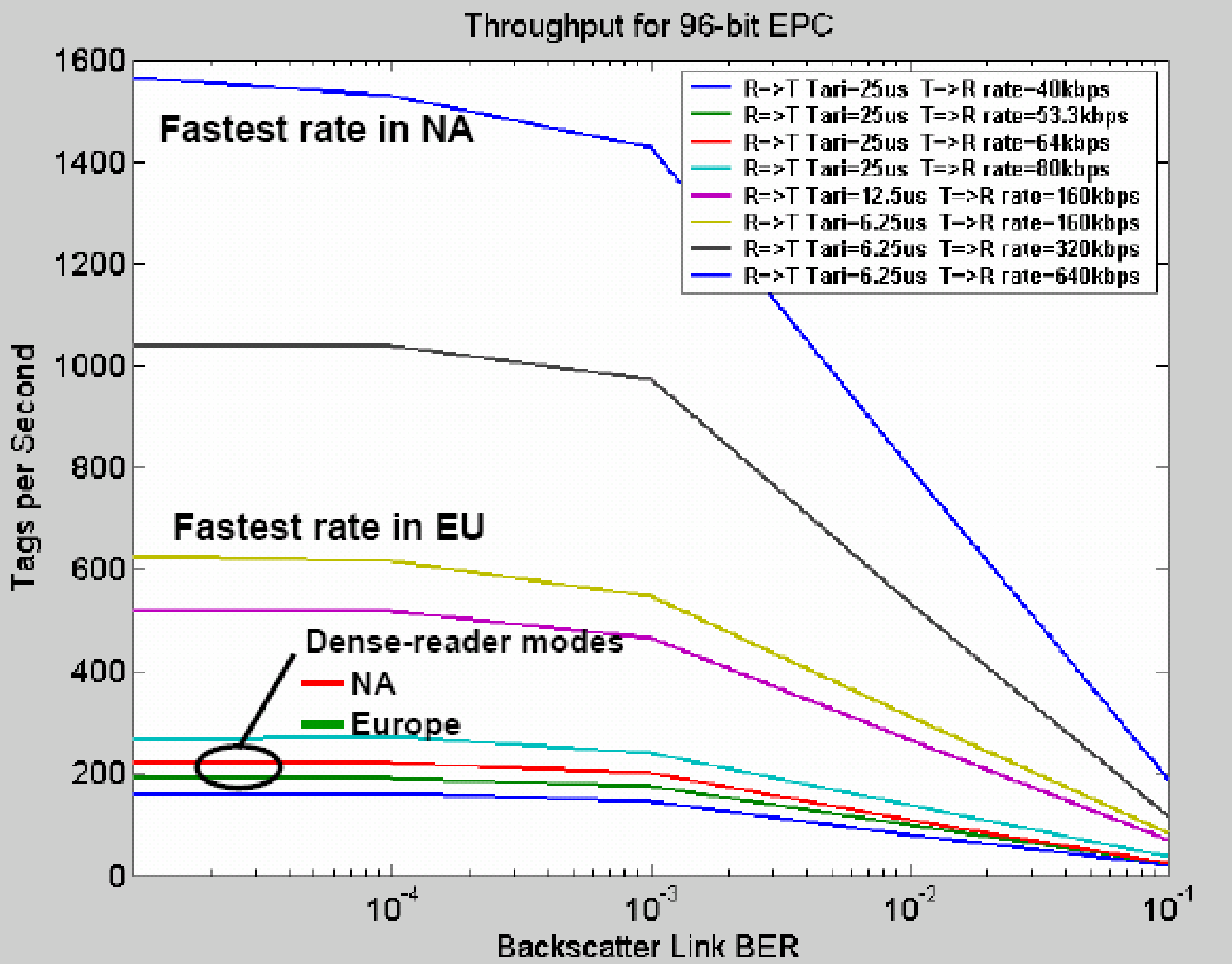


R2T Pass Loss
Tag Available Power
Tag Reflected Power @ R with RCS=4cm2
Tag Reflected Power @ R with RCS=16cm2
Tag Reflected Power @ R with ATNAGain=6dBi & 20% Power Presumption
Tag Reflected Power @ R with Spectrum Analyzer ATNAGain=0dBi & 20% Power Presumption
Measured Tag Reflected Power @ R
Power (dBm)
Link Distance (m)
20
10
0
-10
-20
-30
-40
-50
-60
-70
-80
0
1
2
3
4
5
6
7
8
9
10

Singulate Rate

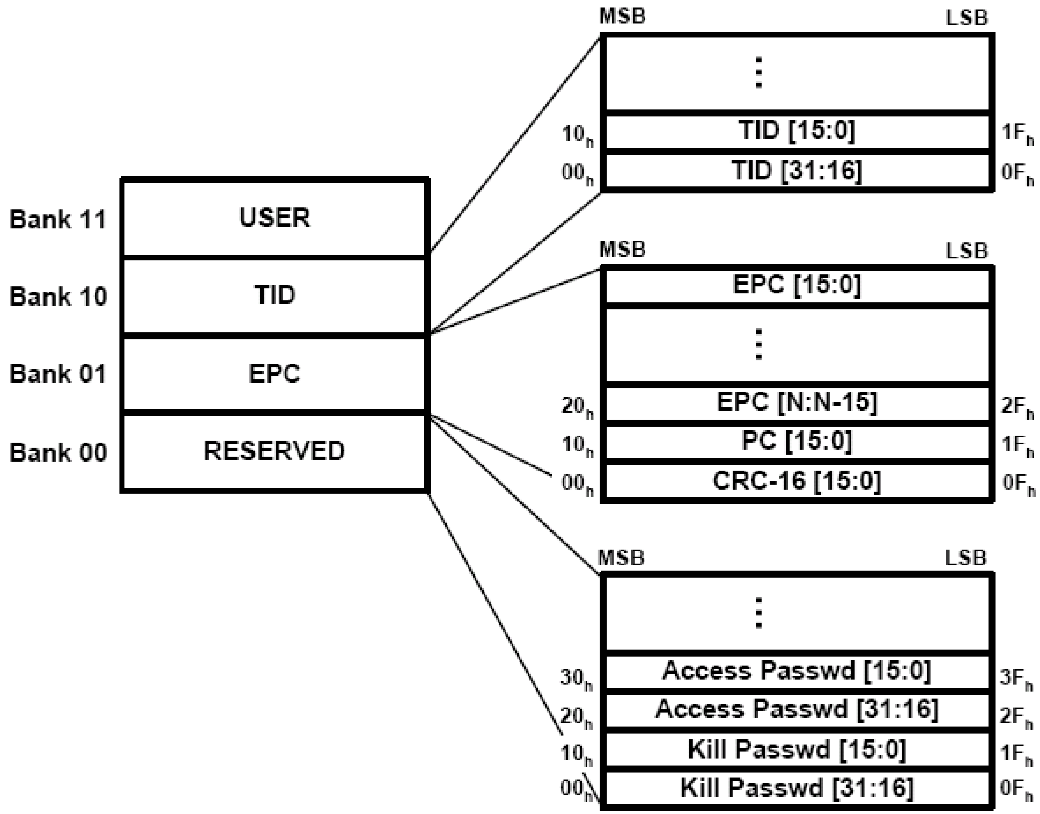
- **Number of tags being singulated in one second**
- **Determined by**
  - Down link speed
  - Up link speed
  - Channel bit error rate
  - Anti-collision algorithm
  - Total number of tags being powered

Singulate Rate

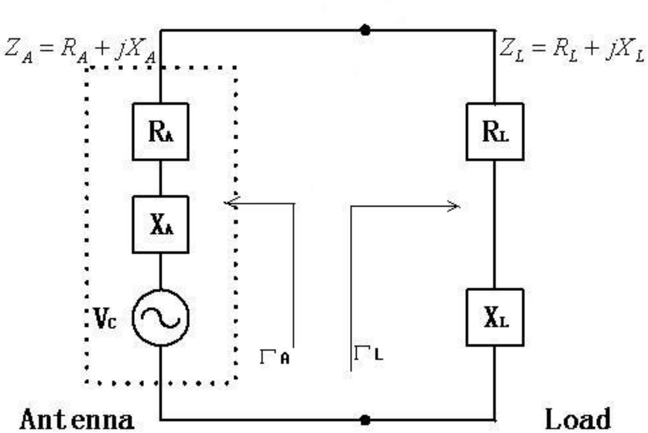


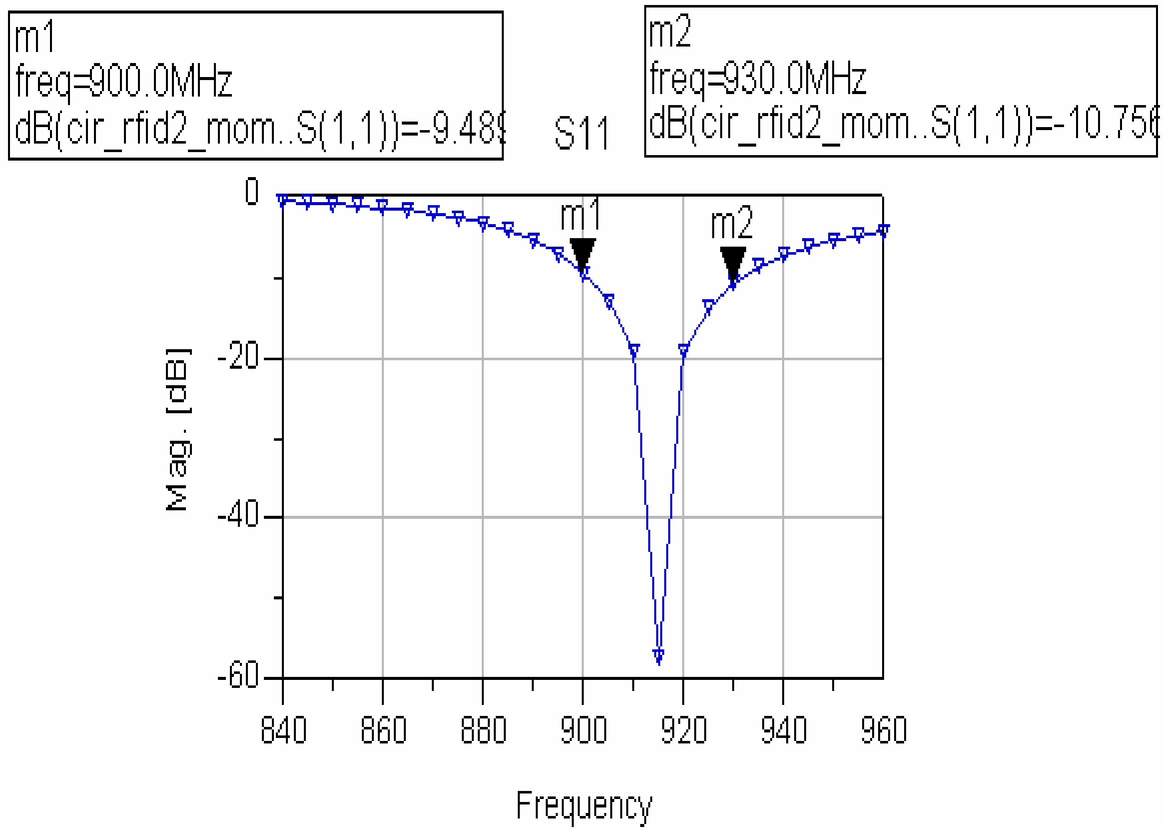
Memory

- **Memory type**
  - Read only
  - Write once
  - Rewritable
- **Memory size**
- **Data retention and endurance for non-volatile memory**
- **Memory organization**



Bandwidth


$Z_A = R_A + jX_A$
$Z_L = R_L + jX_L$
RA
XA
Vc
RL
XL
ΓA
ΓL
Antenna
Load


m1
freq=900.0MHz
dB(cir_rfid2_mom..S(1,1))=-9.48
S11
m2
freq=930.0MHz
dB(cir_rfid2_mom..S(1,1))=-10.75
Mag. [dB]
0
-20
-40
-60
m1
m2
840
860
880
900
920
940
960
Frequency

Tag structure

- **Antenna**
  - Dipole
  - Patch
  - Fractional
- **Chip**
  - Rectifier
  - Regulator
  - Demodulator
  - Controller
  - Memory
  - Modulator

Antenna Design Specification

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The antenna must :

- ✓ be small enough to be attached to the required object
- ✓ have omni-directional or hemispherical coverage
- ✓ must provide maximum possible signal to the ASIC
- ✓ have a polarization such as to match the enquiry signal regardless of the physical orientation of the protected object.
- ✓ be robust
- ✓ be very cheap

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The major considerations

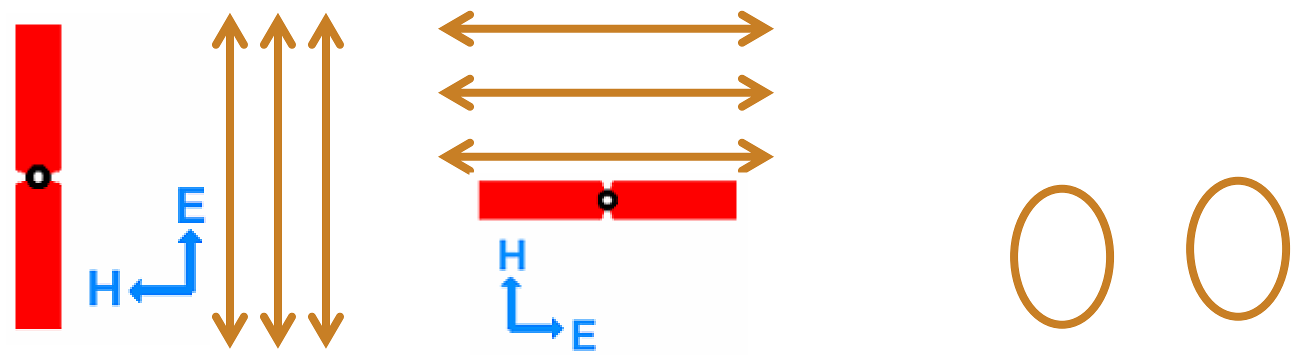
- The type of antenna
- Its impedance
- RF performance when applied to the object
- RF performance when the object has other structures around it

Possible Antenna

Antenna	Pattern Type	Free Space Bandwidth (%)	Dimensions (wavelengths)	Impedance (ohms)
Dipole	**Omni directional**	**10-15**	**0.5**	**50-80**
Folded Dipole	**Omni directional**	**15-20**	**0.5 by 0.05**	**100-300**
Printed Dipole	**Directional**	**10-15**	**0.5 by 0.5 by 0.1**	**50-100**
Printed patch	**Directional**	**2-3**	**0.5 by 0.5**	**30-100**
Log-spiral	**Directional**	**100**	**0.3 high by 0.25 diam base**	**50-100**

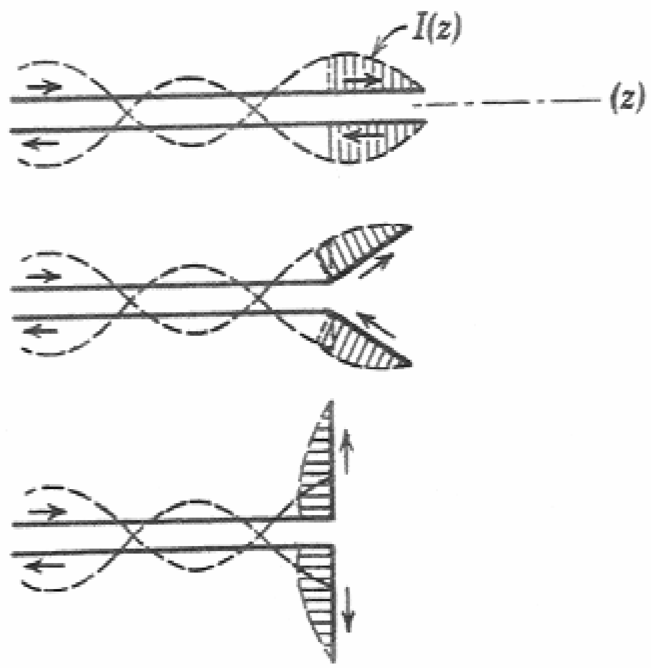
Tag Orientation and polarization

- **Linear polarization**
  - The tag's antenna must be in the same orientation as the reader antenna to be able to receive maximum energy
- **Non-linear polarization**
  - Relative angle of tag and reader antenna will result different read range
- **Circular polarization**
  - Allows any tag antenna orientation



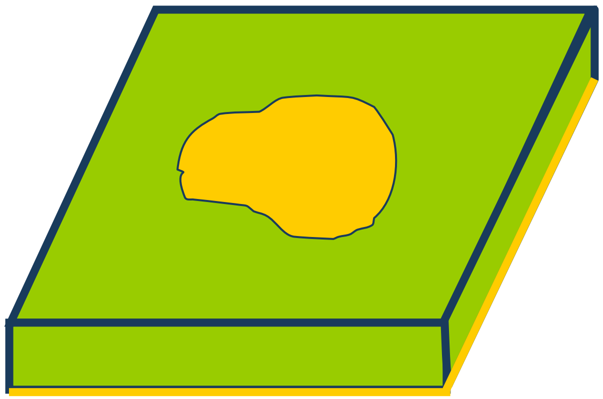
Dipole Antennas

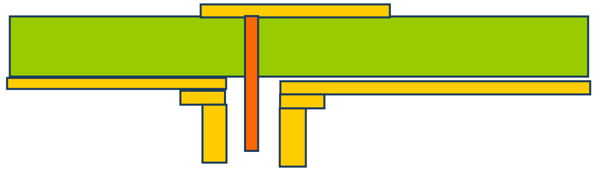
- Wire dipoles can be more costly than printed dipoles due to manufacturing processes, so for very low cost applications, printed dipoles may be preferred. It is important to reduce ohmic losses in the antenna, so generally, the dipole width should be relatively wide. The material used should also have a low resistivity and a high quality factor.
- Though they provide more efficient power transfer, half-wave dipoles, by virtue of their resonance, can have a relatively narrow bandwidth. Bandwidth can be increased, though, by increasing the widths, or effective diameters



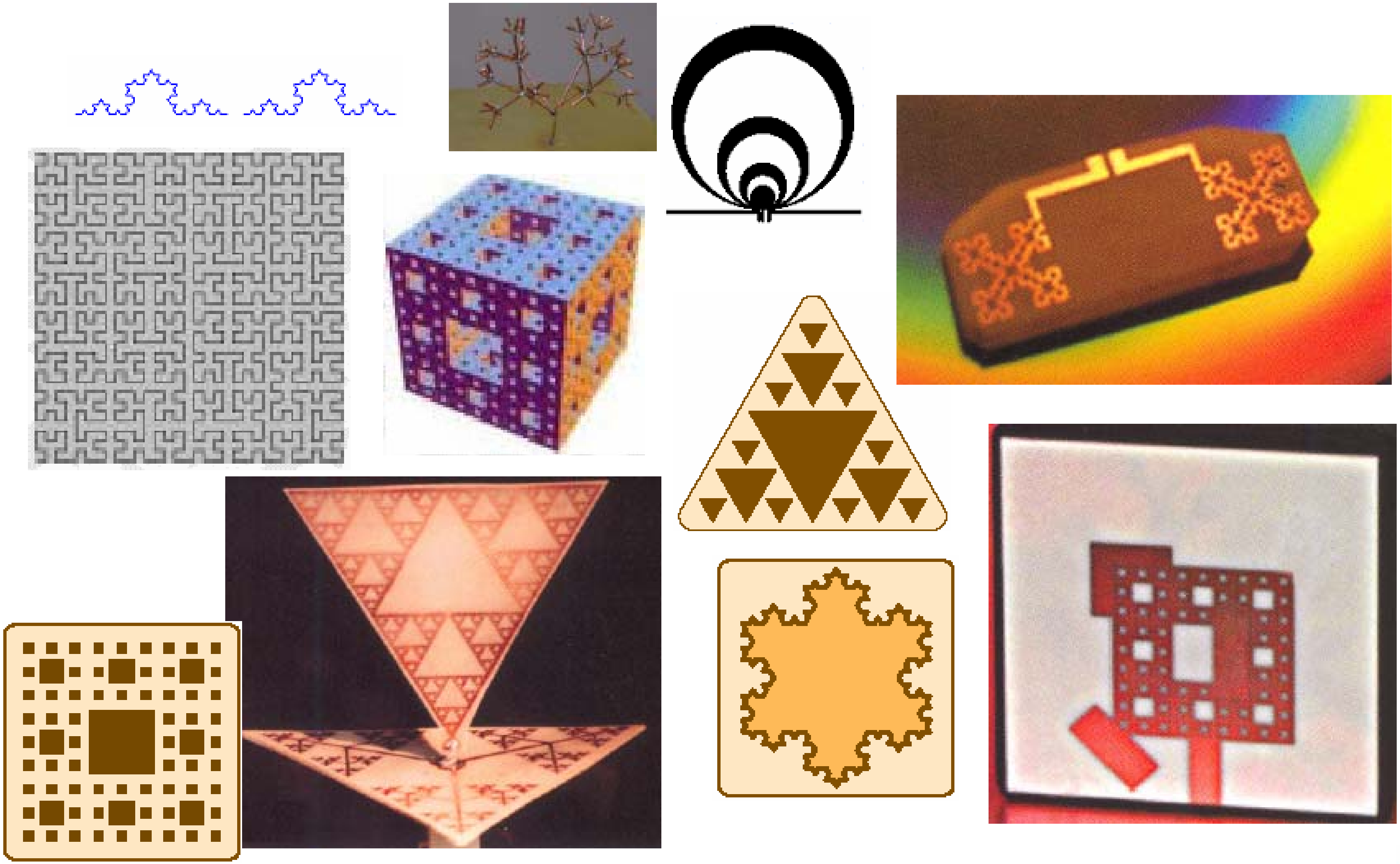
Microstrip Patch Antennas

- Microstrip patch antennas are planar antennas that are typically printed directly onto a dielectric circuit board. They consist of three layers: the conductive patch on top, a conductive ground plane on the bottom, and a dielectric material in between. The patch is usually some appreciable fraction of a wavelength in dimensions, while the ground plane extends further in dimensions. The dielectric material is typically some printed circuit board material – often, PTFE, or Alumina. Because they have a ground plane, their radiation pattern is limited to the area above the ground plane.
- Microstrip patch antennas can be costlier than printed dipoles, as they require the ground plane and dielectric material. Printed dipoles of course need a substrate, yet the substrate material does not have as great as an affect on performance as does the patch dielectric.
- In general, performance offered by dipoles and patch antennas can be comparable, and their use in many ways depends on the application, the operating frequency, and the cost of materials and manufacture.



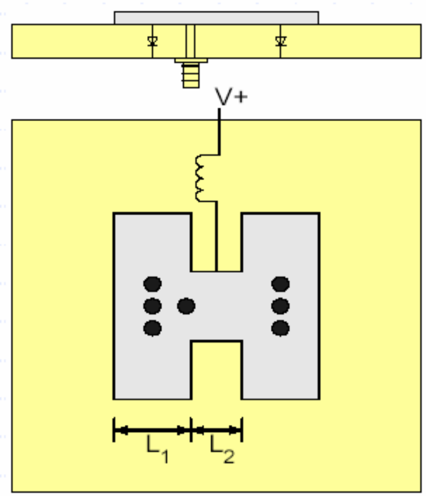


Fractal Antenna



Other Antenna

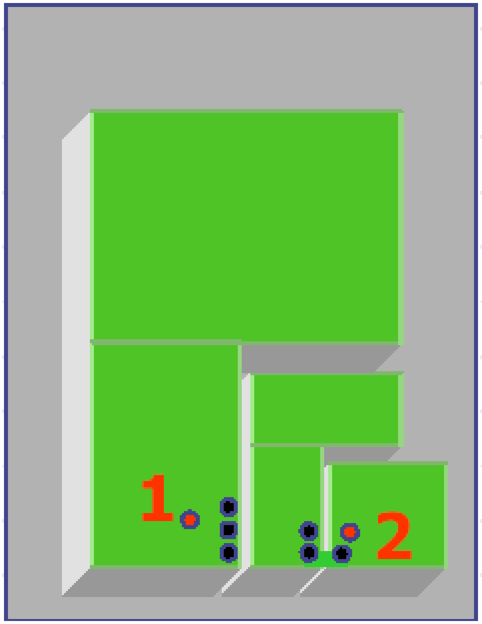
- Multi-band and multi-function antennas
- Pattern diversity antennas



Pattern diversity h-patch

Switches on: monopolar mode

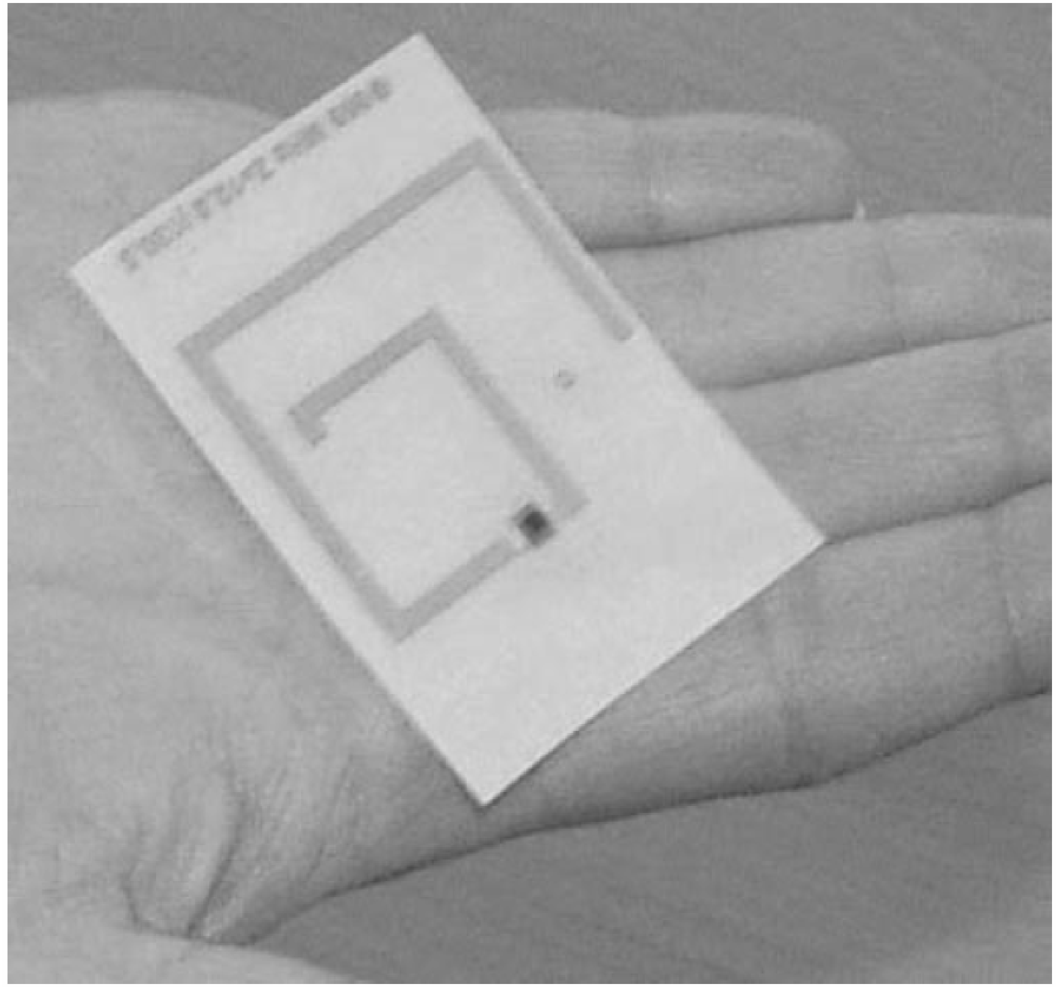
Switches off:CP patch mode



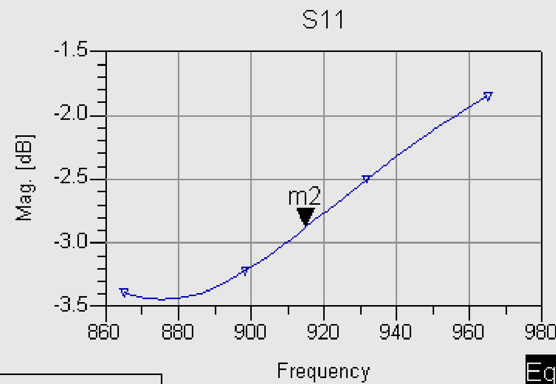
Triple band PIFA

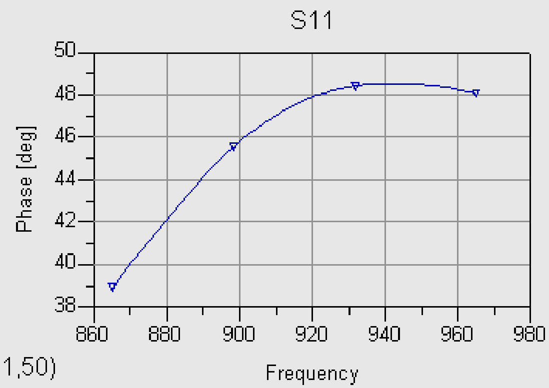
Dual Band Antenna

Operation Frequency:
865MHz and 2.45GHz



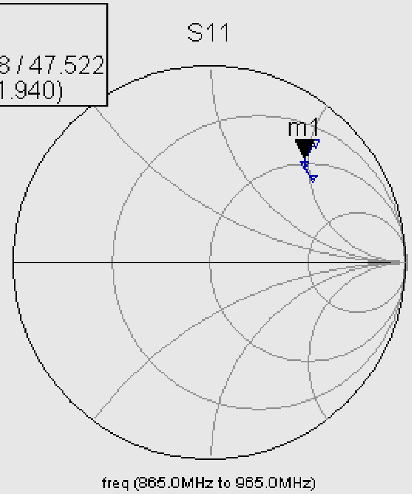
Antenna Simulation Tools





m2
freq=915.0MHz
dB(kk_try_mom_a..S(1,1))=-2.880

m1
freq=915.0MHz
kk_try_mom_a..S(1,1)=0.718 / 47.522
impedance = Z0 * (0.888 + j1.940)



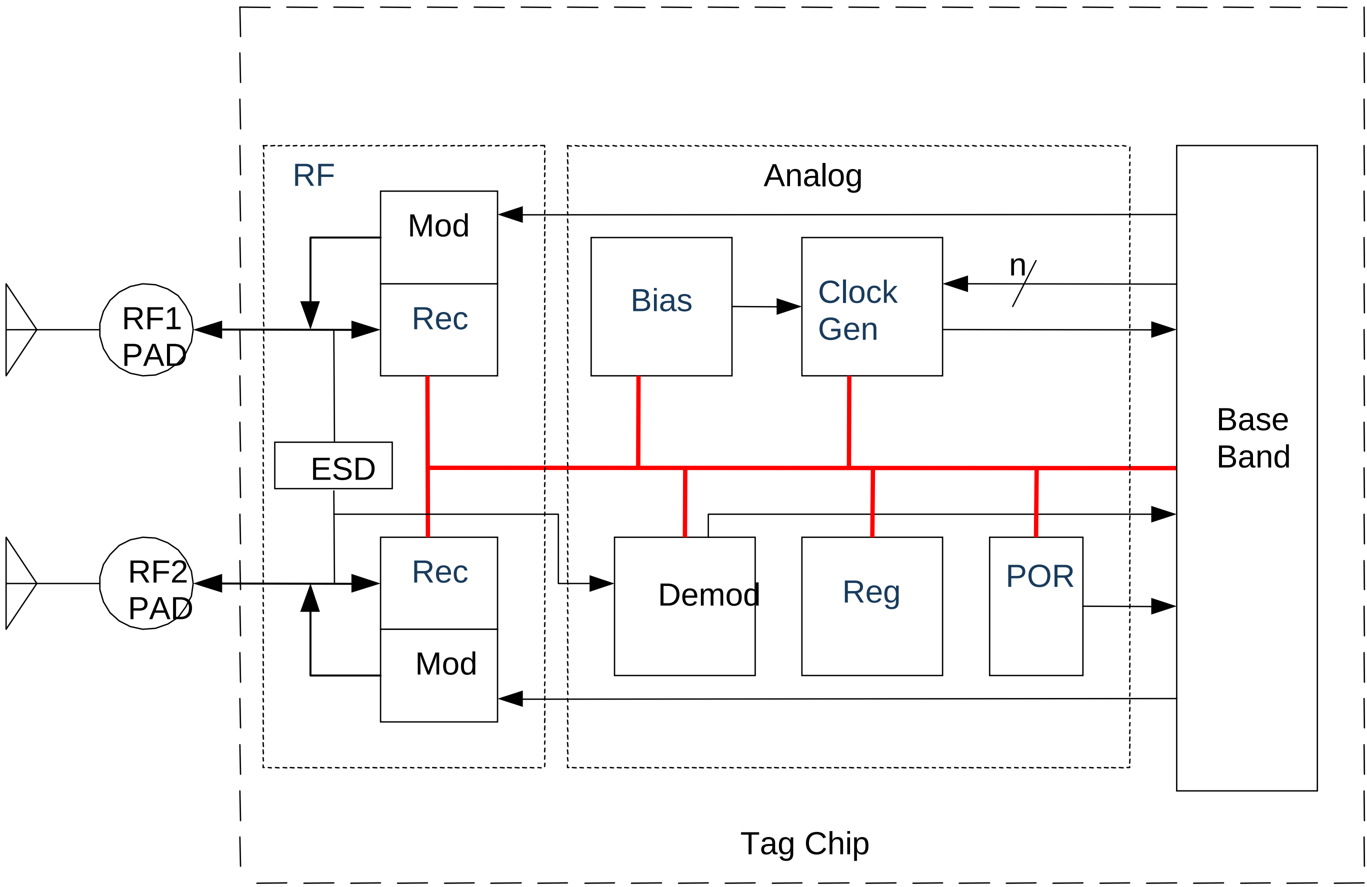
Eqn z=zin(S11,50)

Eqn s=(zin(S11,50)-50)/(zin(S11,50)+50)

freq	z	s
865.0 MHz	66.878 + j105.000	0.677 / 38.933
867.1 MHz	66.014 + j104.000	0.675 / 39.372
869.2 MHz	65.115 + j103.070	0.674 / 39.817
871.3 MHz	64.183 + j102.211	0.673 / 40.267
873.3 MHz	63.231 + j101.421	0.673 / 40.717
875.4 MHz	62.257 + j100.700	0.673 / 41.167
877.5 MHz	61.270 + j100.043	0.673 / 41.614
879.6 MHz	60.271 + j99.453	0.673 / 42.056
881.7 MHz	59.267 + j98.924	0.674 / 42.492
883.8 MHz	58.262 + j98.456	0.675 / 42.919
885.8 MHz	57.256 + j98.045	0.677 / 43.336
887.9 MHz	56.257 + j97.692	0.678 / 43.740
890.0 MHz	55.261 + j97.391	0.680 / 44.132
892.1 MHz	54.275 + j97.138	0.682 / 44.510
894.2 MHz	53.300 + j96.935	0.685 / 44.871
896.3 MHz	52.337 + j96.776	0.687 / 45.216
898.3 MHz	51.389 + j96.661	0.690 / 45.544
900.4 MHz	50.455 + j96.585	0.693 / 45.855
902.5 MHz	49.537 + j96.549	0.696 / 46.148
904.6 MHz	48.638 + j96.548	0.700 / 46.422
906.7 MHz	47.756 + j96.580	0.703 / 46.678
908.8 MHz	46.891 + j96.642	0.707 / 46.916
910.8 MHz	46.046 + j96.736	0.710 / 47.136
912.9 MHz	45.219 + j96.857	0.714 / 47.337
915.0 MHz	44.410 + j97.003	0.718 / 47.522
917.1 MHz	43.622 + j97.173	0.722 / 47.689
919.2 MHz	42.853 + j97.366	0.726 / 47.839
921.3 MHz	42.102 + j97.579	0.730 / 47.973

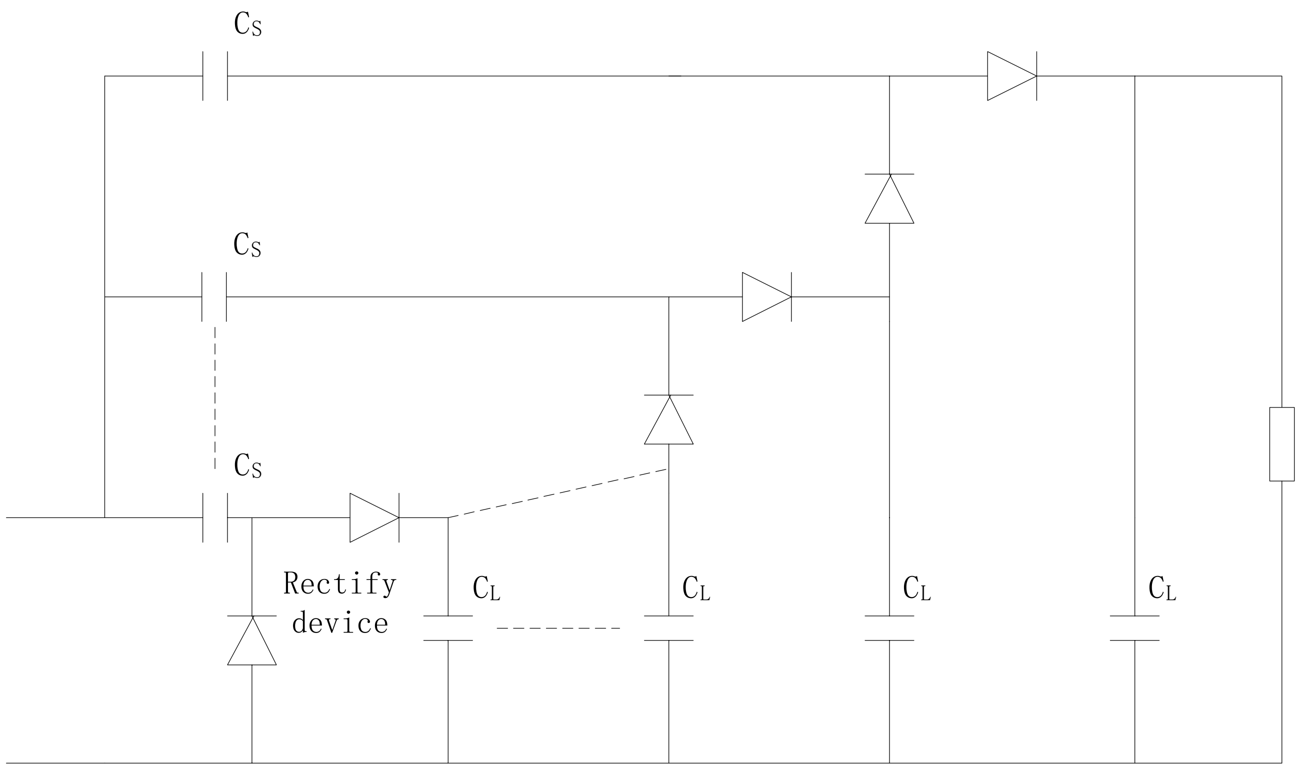
Tag structure

- **Chip**
  - Rectifier
  - Regulator
  - Clock
  - Bias
  - Demodulator
  - Controller
  - Memory
  - Modulator



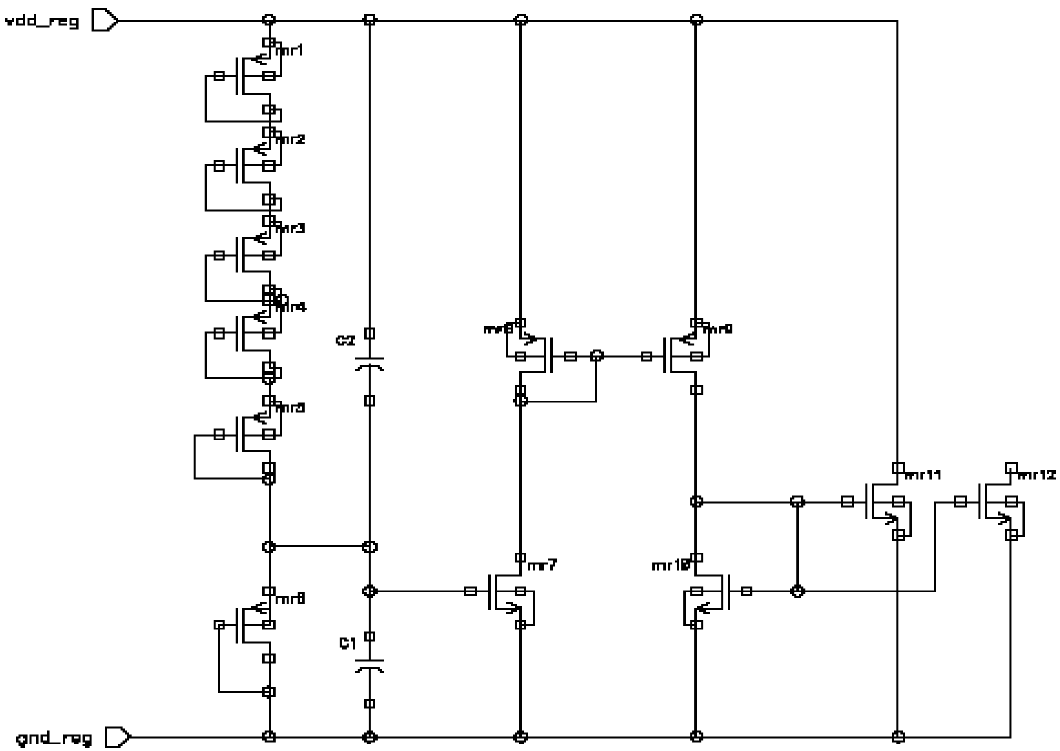
Rectifier (Charge Pump)

- **Convert RF energy to DC**



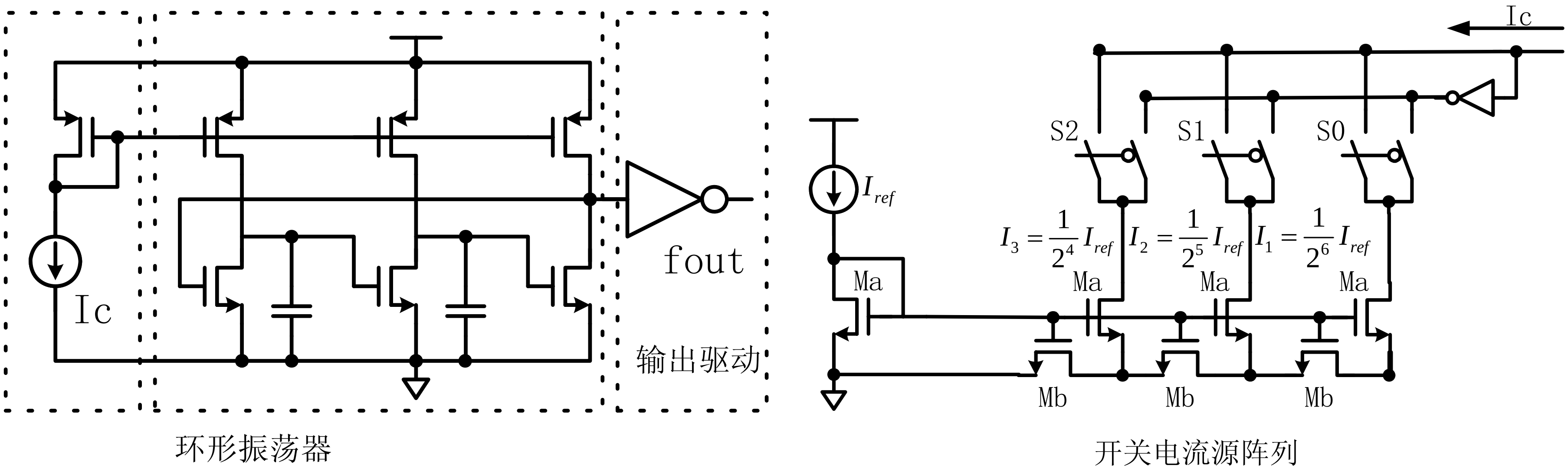
Regulator

- **Stabilize DC power for analog and digital circuits**



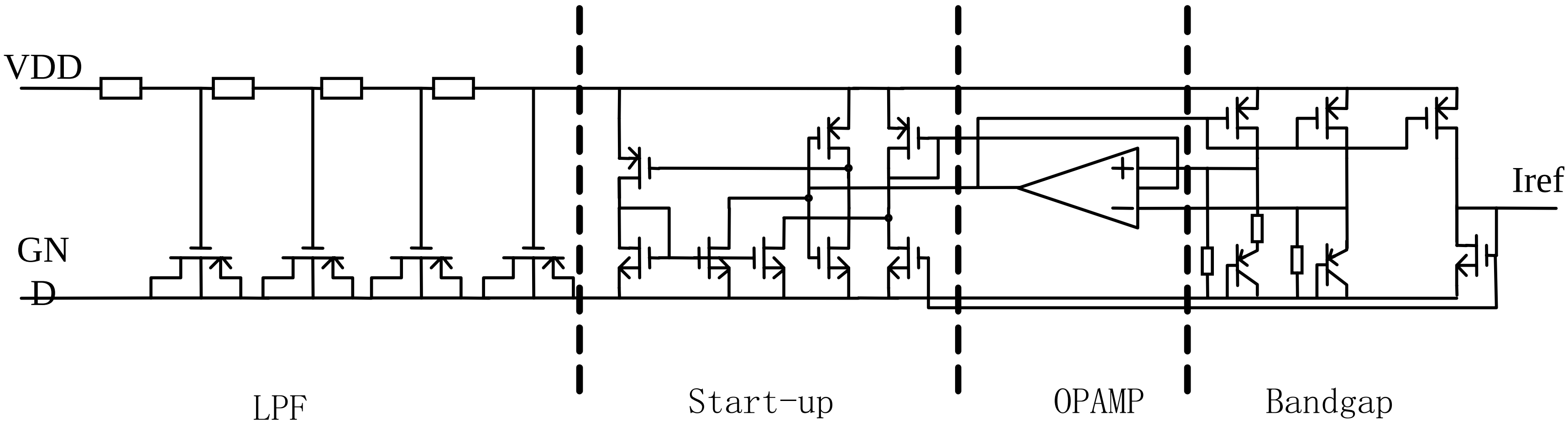
Clock Generator

- **Stable clock. Frequency trimming is needed**



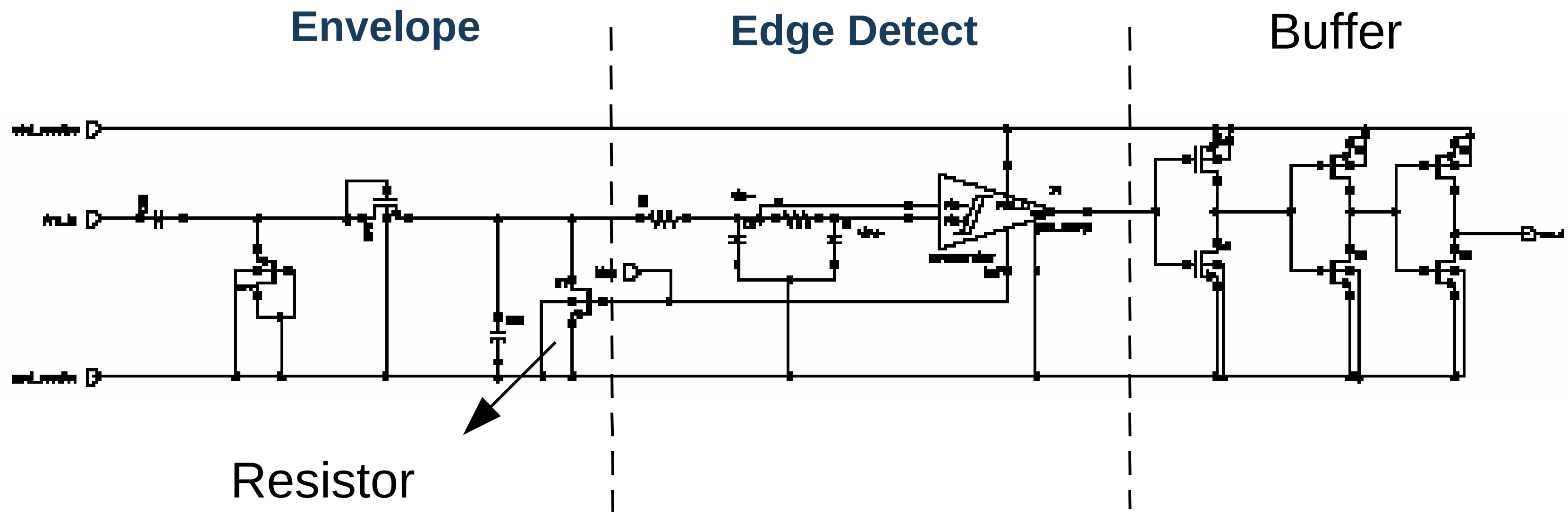
Bandgap Reference

- **Provide a temperature insensitive reference**



Demodulator

- **Demodulate reader to tag link**



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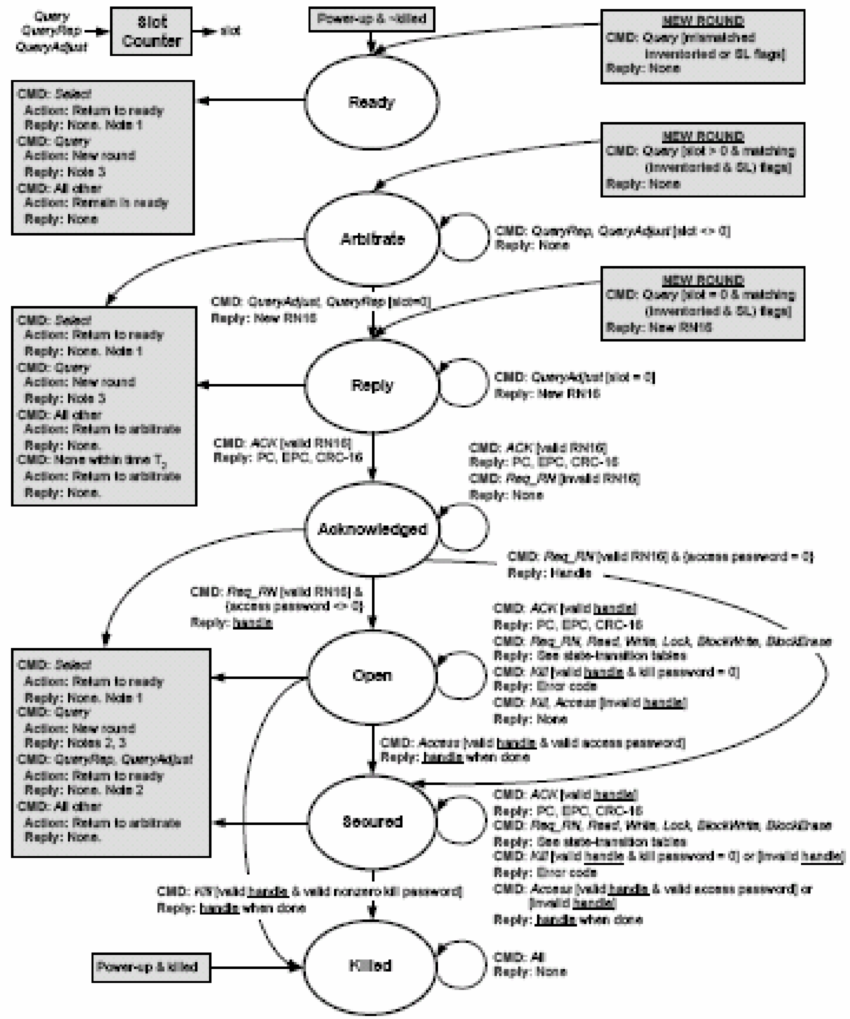
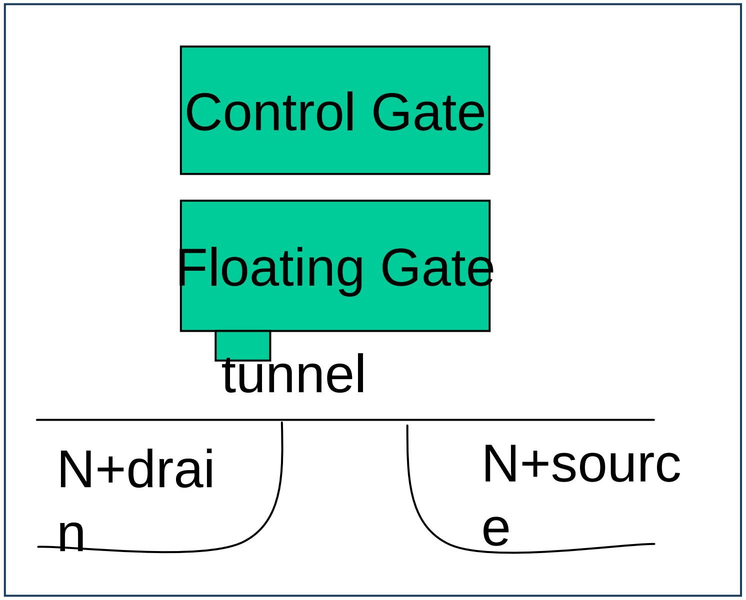
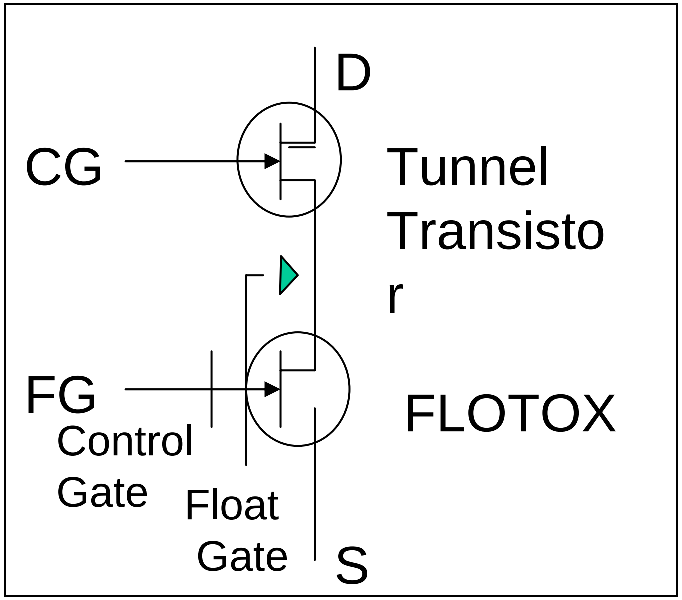
- **Base band digital state machine**



Memory

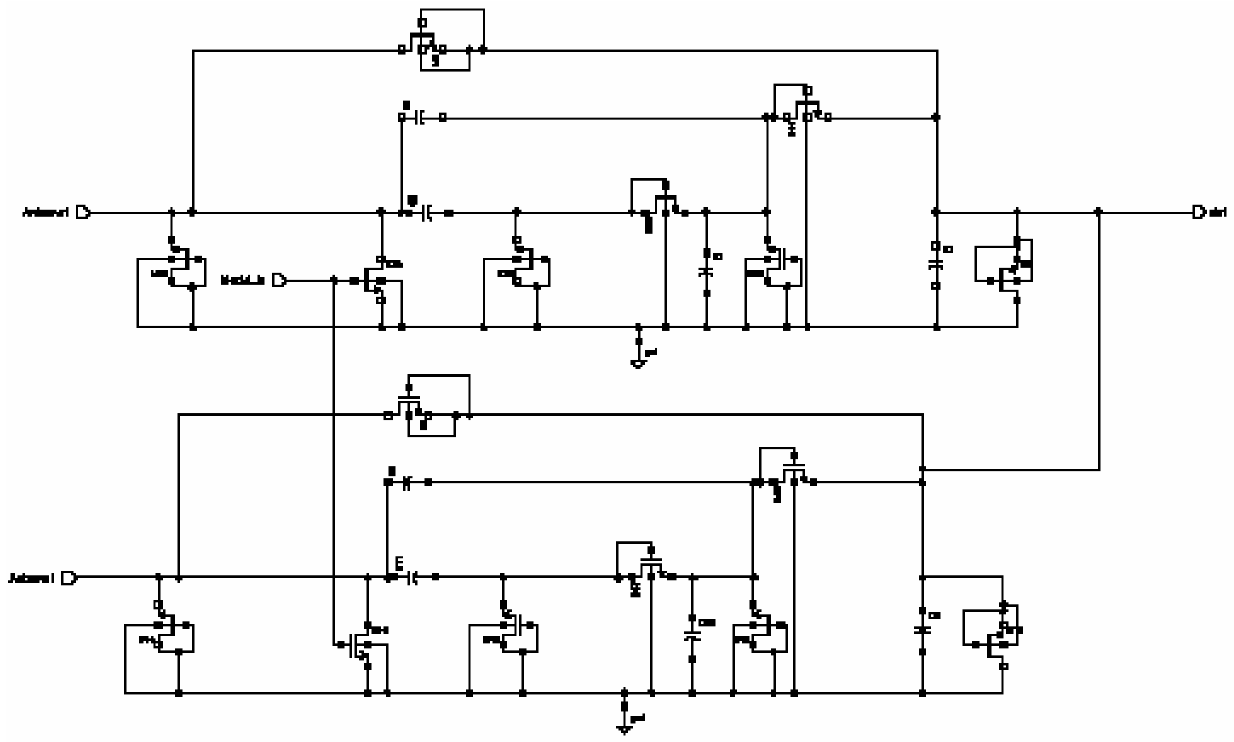
- **OPT or EEPROM is used**





Modulator

- **Switching the impedance between antenna terminals**



Rec,ESD,Modulator

Gen 2 Tag

- **Overview**
- **Features**
- **Technology**
  - Modulation and demodulation
  - Memory
  - Anti-collision
  - State machine
  - Security

History

- **ISO/IEC 18000**
- **ISO/IEC 18000-6**
  - 18000-6A –Supertag, EM
  - 18000-6B –Intermec, Philips

  - Published standard dated 2004-08-31
- **EPC**
  - Originated from Auto-ID center (MIT, etc.) from 1999
  - Transferred to EAN/UCC in 2003
  - Class 0 –Matrics
  - Class 1 –Alien Technology
  - EPC C1 Gen2 ratified in Oct. 2004
- **Merge EPC C1 Gen2 into ISO/IEC 18000-6 as Type C**

Class 0, Class 1 and Gen2

Features	Class 0	Class 1	Gen2
WW regulatory compliance		√	√
Memory access control & locking	√		√
Optional user memory			√
Multi-source availability	√		√
Reads > 500 tags / second	√		√
Kill security			√
Low cost	√		√
Dense-reader operation		√	√
Field-rewriteable memory			√
Industry certification plan			√

Comparison of Type A, B and C

Parameter	Type A	Type B	Type C
Forward link encoding	PIE	Manchester	Uniquely decodeable PIE
Modulation index	27% to 100%	18% or 100%	80% to 100%
Data rate (Interrogator -> Transponder)	33 kbps	10 or 40 kbps	26.7 kbps to 128 kbps
Return link encoding	FM0	FM0	FM0, Miller sub-carrier
Data rate (Transponder -> Interrogator)	40kbps	40kbps	40kbps~640kbps for FM0 5kbps~320kkbps for Miller modulated sub-carrier
Collision arbitration	ALOHA	Binary Tree	slotted random anti-collision
Tag unique identifier	64 bits (40 bit SUID)	64 bits	Variable: minimum 16 bits, maximum 496 bits
Memory addressing	Blocks up to 256 bits	Byte blocks, 1,2,3 or 4 byte writes	Bit, word (16-bit), or vendor-defined block boundaries (depending on command)
Error detection, forward link	5 bit CRC for all commands (with an additional 16-bit CRC appended for all long commands)	16-bit CRC	16-bit CRC, except a 5-bit CRC for the *Query* command
Error detection, return link	16-bit CRC	16-bit CRC	16-bit CRC, except no error check for RN16
Collision arbitration linearity	Up to 250 tags	Up to 2^{256}	Linear up to 2^{15} tags in the interrogator's RF field; above that number, N log(N) for tags with unique UIIs

Type C Advantage

- **ISO 18000-6 Type C/EPC C1 Gen2 was developed by more than 40 companies**
  - First time include technology suppliers and end-users
  - Cross-vendor compatibility
  - World wide interoperability
  - Improvements in performance, cost and reliability
  - Expandable, can build future generations upon it

Key qualities of Type C

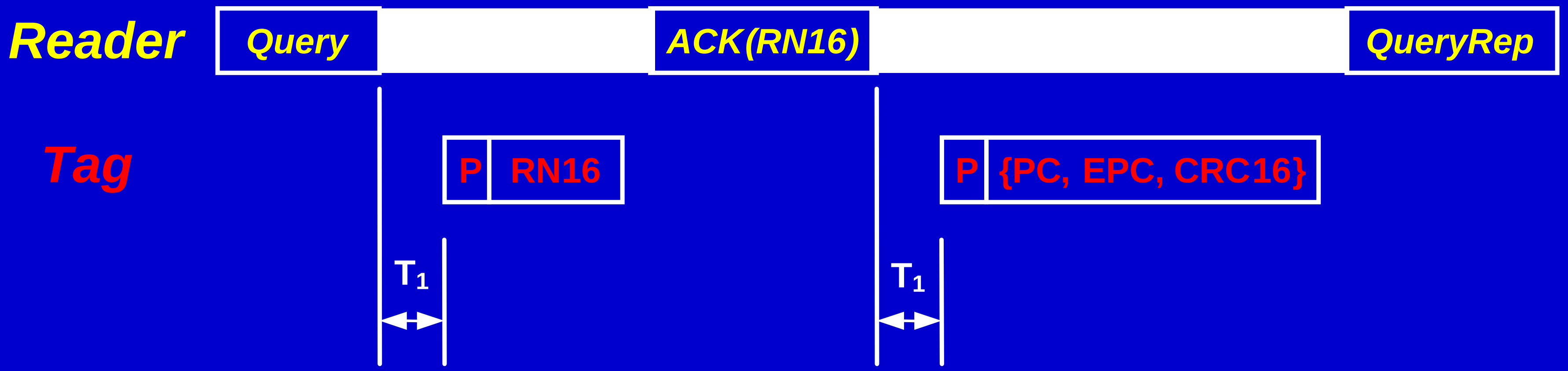
- **Superior Tag population management**
  - Trough Select->Inventory->Access
  - First time introduce sessions
- **Dense-reader mode**
  - Allows > 50 simultaneously operating readers
  - Eliminates need for listen-before-talk
- **Fast and Variable Data rates**
  - Up to 128Kbps (I->T) and 640Kbps(T->I)
  - Flexible to adapt to different applications
- **Kill password**
  - Address privacy concerns while preventing unauthorized kill
- **Robust Signaling**
  - < 1 ghost read / year

Session boost productivity

- ***RFID tag to multiple readers: "Take a number"***
- **Up to 4 sessions**
  - Up to 4 readers can access same tag population via *time-interleaving*
- **Some scenarios**
  - Warehouse
    - Multiple Reader access same tag via time-interleaving
    - *“Can I talk to you for a while?”*
  - Conveyor belt
    - Divide the tag population and conquer
    - *“Get a number and use your assigned entrance”*

No Ghost Reads

- Probability of a ghost read in Type C is << 1/year



Tags must pass 8 checks to be "valid"

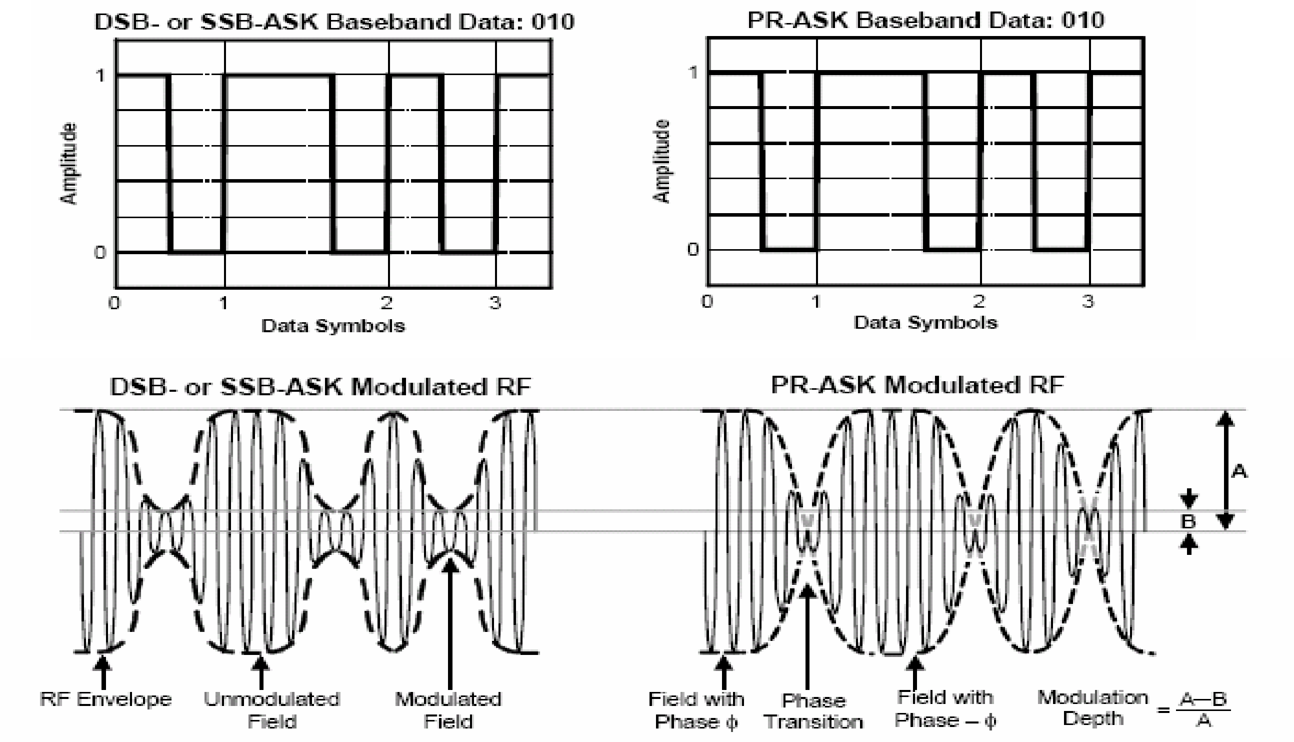
T_1 Timing → Preamble validation → 16-bit path metric → T_1 Timing → Preamble validation → 128-bit path metric → PC length check → CRC16 validation

Type C transponder chip

- **A spec. is not a solution in and of itself**
- **Compliance does matter, so does performance**
- **Key technologies of transponder chip**
  - Low dead-zone high efficient rectifier
  - Low cost, low power consumption non-volatile memory
  - Low power digital circuit
  - Interference Rejecting
  - Cost and performance

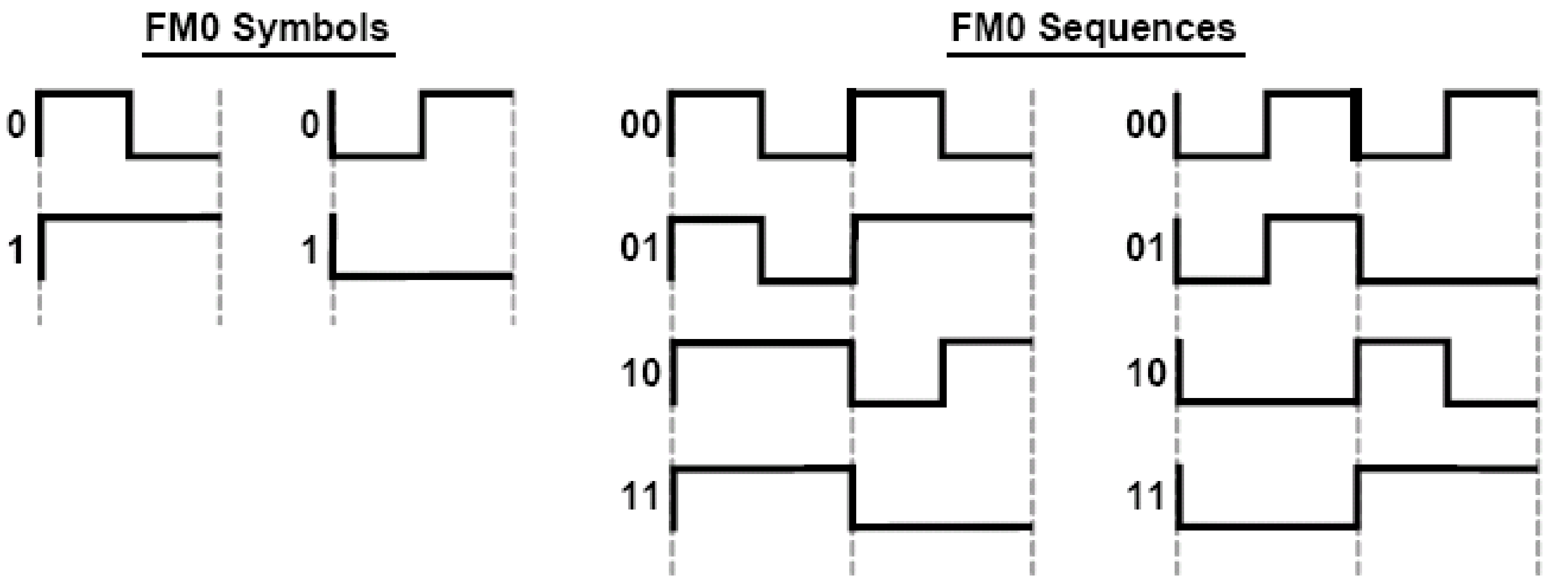
Reader to Tag Modulation

- Speed: 40K to 160K
- Modulation: DSB-ASK, SSB-ASK, PR-ASK
- Encoding: PIE



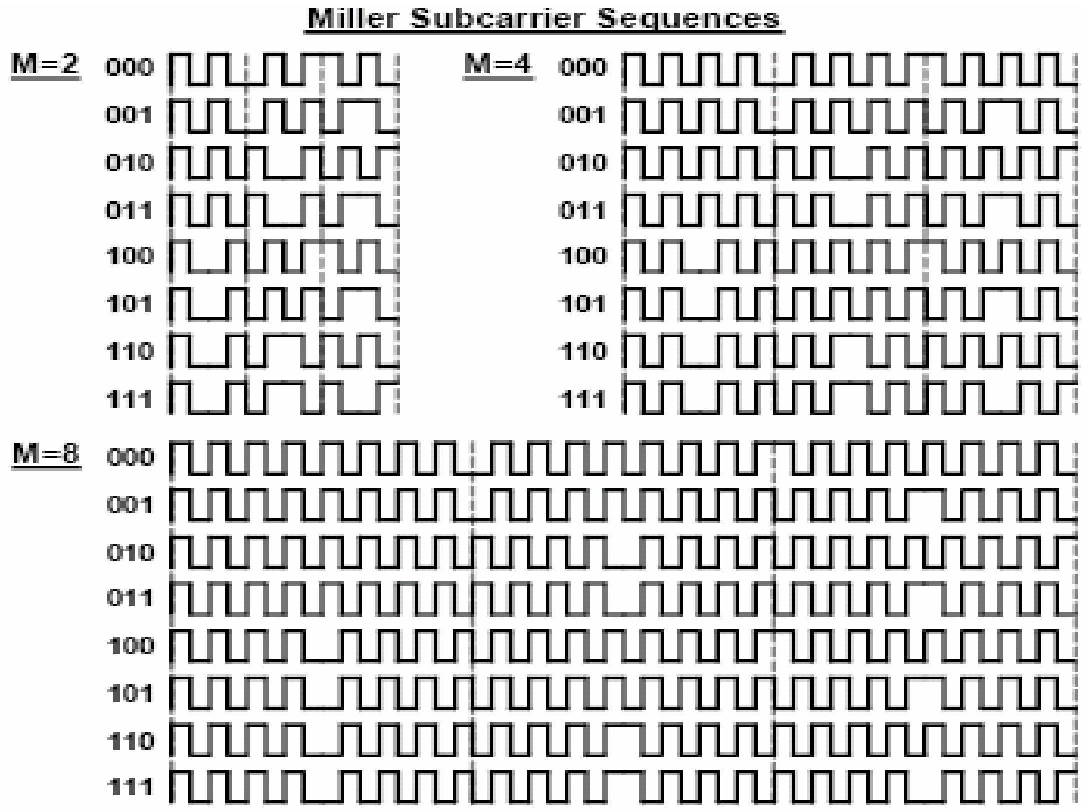
Tag to Reader Modulation

- Backscatter: ASK or PSK
- FM0 and Miller sub-carrier

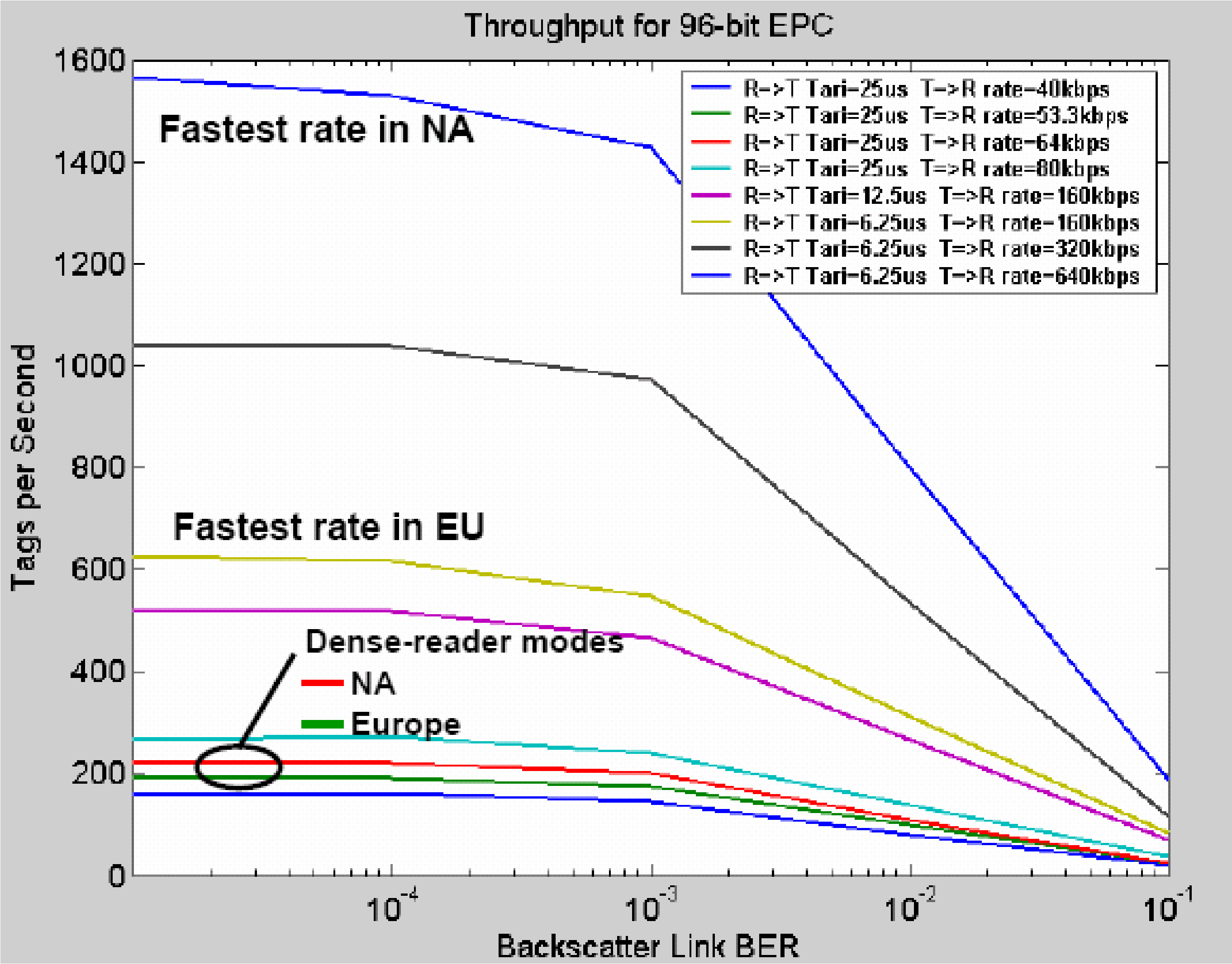


Tag to Reader Modulation

- **Miller subcarrier**

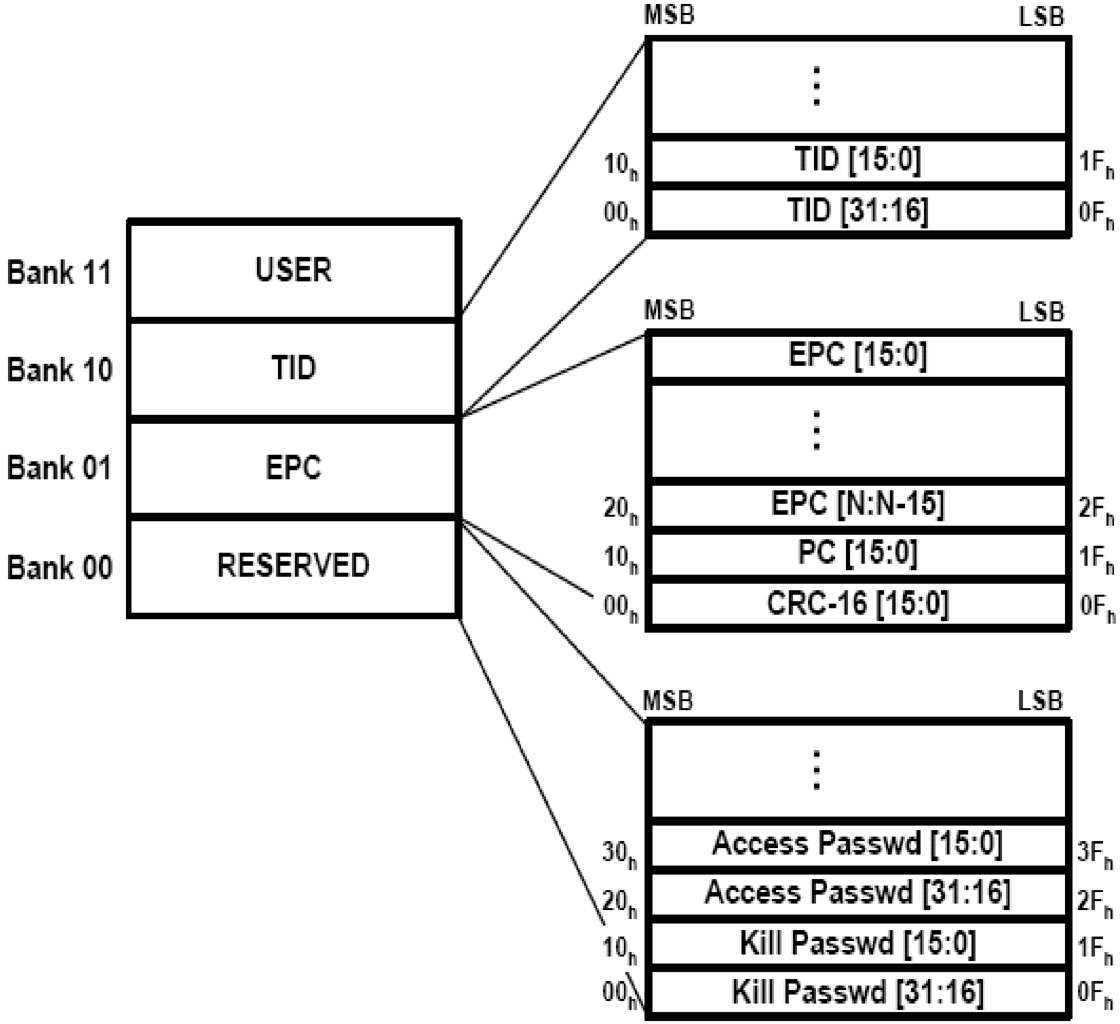


Tag Read Speed



Memory (UII)

- Bits 17_h – $1F_h$: A numbering system identifier (NSI), and which may include an AFI as defined in ISO/IEC 15961.
- If bit 17_h contains a logical 0, then PC bits 18_h – $1F_h$ shall contain an EPCglobal™ Header as defined in the EPC™ Tag Data Standards.
- If bit 17_h contains a logical 1, then PC bits 18_h – $1F_h$ shall contain the entire AFI defined in ISO 15961



Memory (TID)

- TID memory shall contain an 8-bit ISO/IEC 15963 allocation class identifier at memory locations 00h to 07h. TID memory shall contain sufficient identifying information above 07h for an interrogator to uniquely identify the custom commands and/or optional features that a tag supports. This identifying information shall comprise a 12-bit tag mask-designer identifier at memory locations 08h to 13h and a 12-bit tag model number at memory locations 14h to 1Fh. Tags may contain tag- and vendor-specific data (for example, a tag serial number) in TID memory above 1Fh.

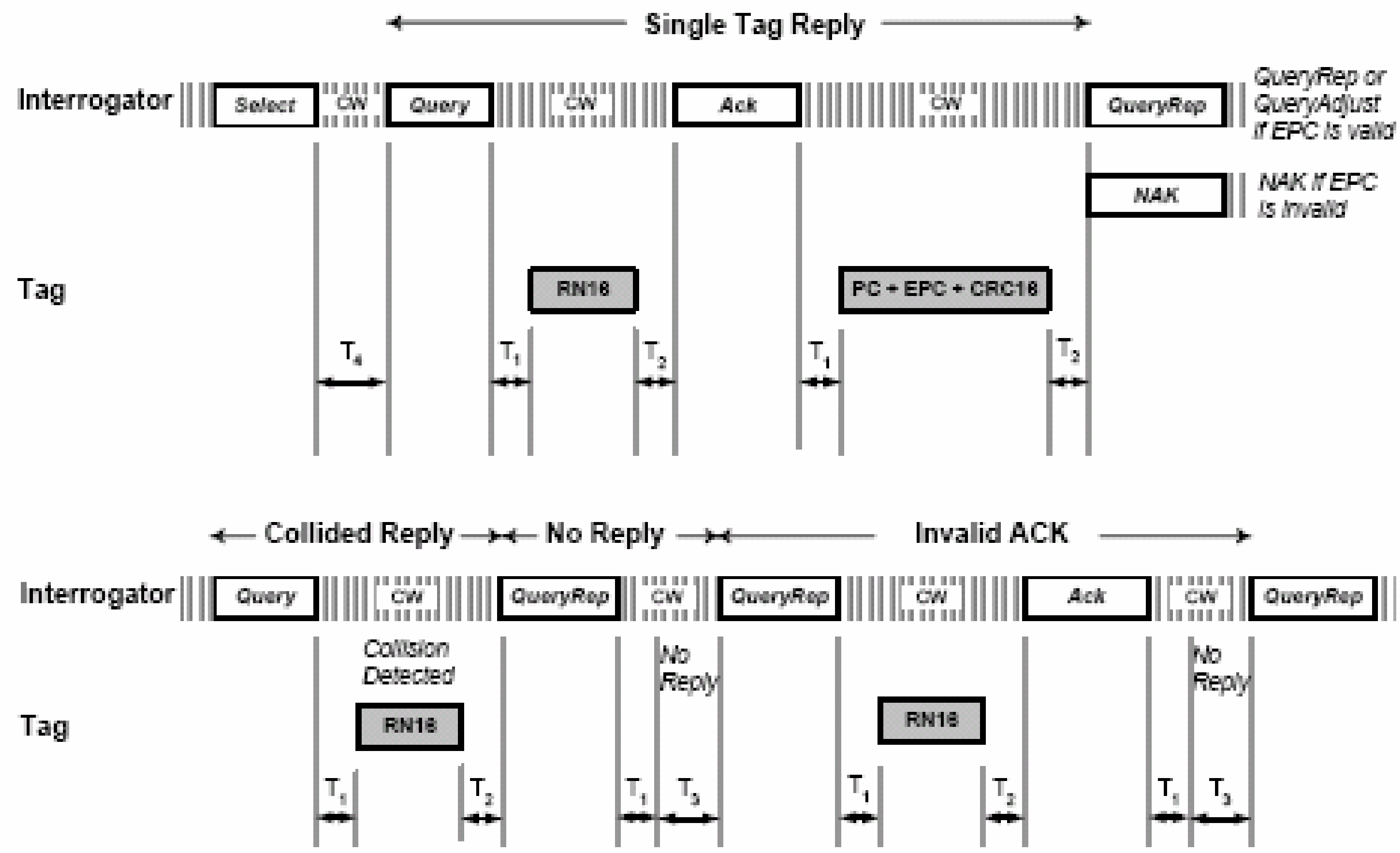
Flexibility on Data Format

- **No obligation on data format**
  - China can control its own identification product number!
- **Type C supports ISO, EPC unique id number**
- **Open door to application specific coding scheme**
- **Provide superior air interface while maintaining flexibility of information on chip**
  - World-wide interoperability
  - Standard means low cost infrastructure
  - System and data security is controllable

Anti-Collision

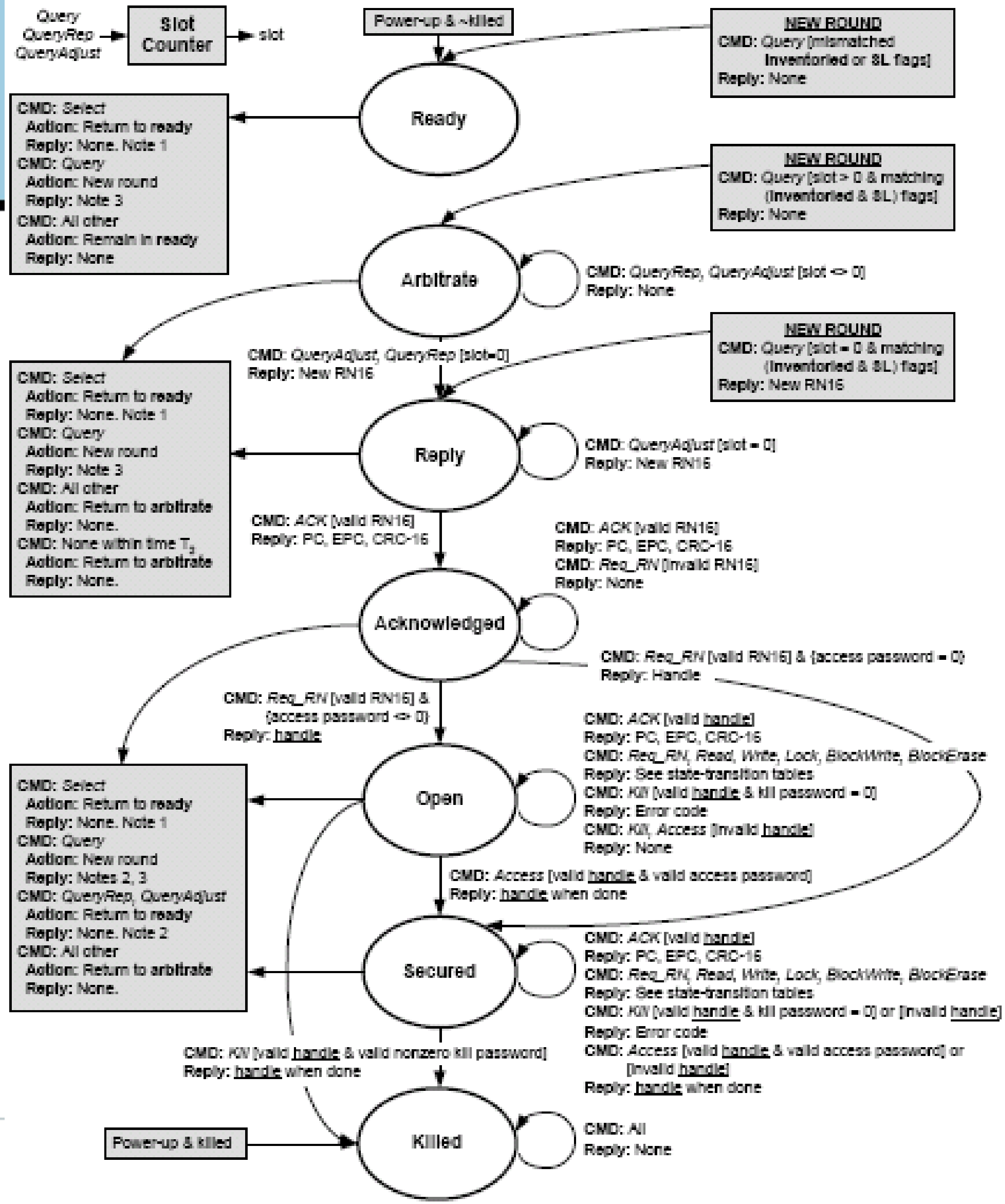
- **Reader issues a Query command with a parameter “Q” (e.g. “5”)**
  - Starts the inventory round
- **Tags load a Q-bit random value into a slot counter**
  - If a tag loads a zero, it then backscatters an RN16 (16-bit random number)
- **If no collision**
  - Readers acknowledges the tag by returning the RN16
  - Acknowledging tag backscatters its EPC
  - Reader issues a QueryRep (sleep) command
  - Tag inverts its inventoried flag and leaves the round
  - All other tags decrement their slot counters
  - If any other tag decrements to zero, it replies with an RN16
- **If collision, reader send NACK (Negative acknowledgment)**
  - Responding tags all put new random number into slot counter
  - Reader sends another QueryRep command and all tags decrement slot counters

Anti-collision Sequence



State Machine

- Ready
- Arbitrate
- Reply
- Acknowledged
- Open
- Secured
- Killed



Security and Privacy

- Reader does not transmit electronic product code(EPC)
- Tag memory can be locked
- Access is password protected
- 32-bit access password
- “Kill" command can permanently disable the Tag
- Protected by 32-bit kill password
- Masked Reader -> Tag communications
- RN16 sent by tag used to mask Reader payload
- Protects data, passwords from eavesdropping

Questions

- **Questions?**
- **Contact:**
  - Hao Min
  - Auto-ID Lab at Fudan
  - Email: hmin@fudan.edu.cn

Acknowledgement

- This lecture note has been summarized from the research work of Auto-ID Labs and other research institutes all over the world. I can’t remember where those slide come from. However, I’d like to thank all professors who create such a good work on those lecture notes. Without those lectures, this slide can’t be finished
- Special thanks to Prof. Chris Diorio, who provided me very detailed analysis on Gen2 tags and readers.